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Alchemic Reality

Moving from augmented to augmentable reality

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Alchemic Reality

Moving from *augmented* to *augmentable* reality

Abstract:
Augmented and virtual reality (AR/VR) is a complex digital medium. The difficulty associated with learning how to build with the medium often blocks new designers and laypeople from fully engaging in the craft. In this thesis the researcher engages with making points of entry through seamful design. A material lens is applied I to understand what affordances exist in AR/VR. AR/VR authorship tools are reviewed to see which affordances are less accessible to laypeople. Finally a prototype tool, Alchemic Reality, is made to explore the role that playful interactions and spontaneity have in AR/VR literacy for new designers and laypeople.

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1. Introduction

This thesis explores making an entry point to the medium of augmented reality for laypeople and new designers. Two experiences were pivotal in discovering and engaging with the issues that face people new to the craft.

My mother is an excellent illustrator and pattern designer. It is a craft she partly learned from her parents, both artists, and mostly gained through hours of practice and hard work in earning her masters degree in fine arts. When she took to designing scrap book kits I was somewhat perplexed. At the time I was in the middle of my own fine arts degree, spending ungodly amounts of hours learning artistic techniques. In my young artist pride I saw scrap booking as a way that people could feel like artists without putting in the required effort.

Yet her view humbled me. She saw scrap booking as an approachable entrypoint for people who otherwise may never explore layout, typography, or papercraft. She explained that everyone needs an entry point that fits them. She said that they needed a chance to have someone help them build something they were proud of.

A whole 15 year design career later, I find even more wisdom in her view than ever. During the second year of my master's program in interaction design, I had the opportunity to create coursework for an introduction to Augmented and Virtual Reality (AR/VR) class. The material was taught over a period of 2 weeks and students were expected to build something in AR/VR. After teaching both bachelor's and master's level students I found AR/VR needs a better entrypoint.



Fig 1: *Meta Quest 3 headset.* From “Meta Quest 3: Next-gen virtual reality headset,” by Meta Platforms, Inc., 2026, Meta Store (https://www.meta.com/se/en/quest/quest-3). Copyright 2026 by Meta Platforms, Inc. Used under fair use for scholarly purposes.

The device available for the class was the Meta Quest 3, Fig 1. This head mounted display (HMD) has been designed with adoption in mind (Öncüler, 2024). But while the device is easier to obtain and use than past iterations, students still ran into approachability issues while designing in the medium. Many of their issues arose with the technical skill required to really utilize the different *materials* at play in AR/VR.

Materials are often thought of in the physical sense of how things are built. A shirt is made of fabric and thread. By understanding the properties of fabric and thread you also understand more about the shirt. Specific properties, like the stretchiness of thread or durability of the fabric, come together to form the shirt. Someone had to understand and act upon the properties of thread and fabric to make the shirt. The properties of a material that one can act upon has been dubbed *affordances* by perceptual psychology (Gibson, 2015, Chapter 8; Norman, 2008). Craftspeople then are those who understand and can manipulate the affordances in their target materials.

Similarly, interaction designers must learn to understand the affordances of digital materials that make up apps and computer programs (W. W. Gaver, 1991; Wiberg, 2018). But a conceptual understanding of a material's affordances aren’t enough. Tools and technical skills are required. To manipulate thread, a seamstress acquires mastery of the needle. Tool choice plays a large role into what skills are needed. A sewing machine with different stitch patterns makes larger projects easier for a new seamstress, but it also governs what stitches can be used. In digital mediums, the tools and technical skills revolve around computer logic and code.

In the course I taught, students were all at different coding skill levels. Two options for tools were given. The game engine Unity (Unity Technologies, 2025) with the Meta development kit

(Meta Platforms ,Inc, 2025) offered advanced access to affordances at the cost of higher technical skills. The immersive app ShapesXR(Shapes Corp, 2025) offered easier building at the cost of a more limited set of affordances.

Watching students engage with their AR/VR projects showed a larger issue facing AR/VR, technical medium literacy. Student groups who had less technical background spent most of their time approaching what kind of project would be achievable with the more approachable tool. Students who had coding backgrounds felt more confident in exploring what problems they could solve. This was outside of prior AR/VR experience and more connected to the skill of understanding computer logic and coding skills.

In the end the students that seemed to take the most pride in their projects were the ones that were able to engage more fully with the medium and explore its many affordances. They engaged with a problem or challenge that was unique to them. For these students it seemed AR/VR was something they could begin to think and play with.

There were several students that expressed frustration. They had huge ideas and a desire to explore so much more of the medium but they spent their entire time fighting tools. This turned them off of AR/VR as they felt it was just not for them. Yet when asked what they would do if they had complete control they offered novel and interesting use cases that connected to their own experience. One must ask then how much AR/VR is held back from reaching its potential because laypeople aren't afforded the ability to build with it.

Similarly the AR/VR field in general is driven by those with enough technical skill to build, or can pay to build, in the problem areas they identify. The skill gap acts as a filter for application cases that favors the minds and paradigm of advanced coders. Domain experts, who happen to be tech skill laypeople, could likely create ground breaking applications if they just knew what affordances AR/VR has and how to apply them.

A 2024 article by Jared Lanier, a big voice for extended reality (XR) since the 80s, revealed an old dream for the craft, “The hope was to make V.R. a place for spontaneous invention. Virtual reality would allow groups of people to ‘play the world into existence’...”(Lanier, 2024) The idea of *playing a world into existence* denotes a playful manipulation of AR/VR. For the laypeople students in my class, technical skill made it far from playful and spontaneous. As tools evolve and hardware gets better one wonders if the future could offer such playful creative access to laypeople. What would it take then to get from the current state of AR/VR creation to the ideal state of *spontaneous invention*?

This thesis takes the idea “playing the world into existence” and “spontaneous invention” in AR/VR from Lanier (Lanier, 2024) and uses it as a means to design gateways for laypeople to create with the medium. The idea is condensed into the phrasing *playful spontaneous creativity*.

Throughout the rest of this thesis I will refer to myself as *the researcher*.

1.1 Research Mission & Research Questions

The overarching mission of this thesis is to move technical skill heavy AR/VR design processes toward the ideal state of *playful spontaneous creativity*. In doing so it seeks a broadening of the new AR/VR designer pool to include people of more varied backgrounds. A prototype was made to help users engage with crafting in the medium with limited need for a coding background. The hope is that these new designers appropriate AR/VR in novel ways and give the medium added meaning and function.

In pursuit of that mission this thesis explores the following research questions:

How do AR/VR authorship tools need to evolve to allow for playful spontaneous creativity amongst new designers and laypeople?

The thesis opens this question up into three subquestions:

R1: If we consider AR/VR to be a medium made of multiple materials to be manipulated – what are the available materials?

R2: How do different AR/VR authorship tools give users access to these materials and which materials are locked behind technical skill walls?

R3: Which design choices in AR/VR authorship tools lead to deeper literacy for laypeople and new designers?

1.2 Resulting Artifact



Fig 2: screenshots of Alchemic reality, a playful augmented reality sketchpad. Images made by the researcher.

The final prototype of this thesis is called Alchemic Reality, seen in Fig 2. The tool acts as a playful sketch pad, letting users explore affordances within AR/VR that are often under-represented in tools aimed at laypeople. Users can sketch with primitive shapes and basic colors to build virtual objects and fuse them together using hand gestures. They can play with how the virtual object reacts with their physical surroundings. They are presented with logic-driven toys to fuse with their creations. The toys are built of smaller logic blocks that take the form of transmutation circles, inspired by portrayals of alchemy in pop culture. These smaller blocks can be accessed by the user to build novel interactions and virtual toys. The spatial data of the physical room has attachments to connect to the logic sketching system, allowing users to playfully code their environment. A short video demonstration is available as Appendix A.

1.2.1 Intended Audience and Test Users

This thesis is written for AR/VR designers to understand the design decisions and research methods used to build Alchemic Reality.

The prototype itself targets interaction design bachelor and master students as its intended audience. Interaction design is a multidisciplinary degree, inviting both engineering and art inclined students. This project is looking to close a gap observed in their learning experience.

A secondary target audience for Alchemic Reality is laypeople who may enjoy AR/VR authorship but lack a suitable entry point.

2. Background and Theory

In order to open up the research area this thesis starts by examining a few ways in which technology platforms can be built to engage with new users and new use cases: Technology Centered Design, Contextual design, Literacy, Seamful design, and Appropriation. It does so through a literary review of these concepts.

Before diving into approaches, this thesis will establish how it views AR/VR. The concept of augmented and virtual reality has older origins than many realize (Cipresso et al., 2018, pp. 2–3). Some taxonomies map the differences based on how much virtual or real world content is displayed (Milgram & Kishino, 1994). One can also look back to one of the original builders of VR devices who saw it as a means to view the virtual world with no distinction of if it was fully virtual immersion (Sutherland, 1968). Others look at augmented reality at the interaction level of real world objects affecting broader digital systems (Mackay, 1998).

This thesis looks at AR/VR on both a visualization and interaction world. It views virtual reality in the old sense of visualizing components of the virtual world while augmented reality is only breached when the real world begins to inform both the user's interactions and the systems calculations. In this way the thesis sees AR and VR as not being mutually exclusive. The acronym AR/VR has been chosen to convey an augmented reality first thinking to the medium, so as to focus on keeping the physical world in mind.

2.1 Technology Centered Design in AR/VR

Large AR/VR tech companies spend a lot of time developing and seeking broader technological capability. Commercial metrics of growth in AR/VR often focus on hardware factors to examine user and industry adoption (Uytven, 2024). In the search for industry relevance, big companies like Meta have developed headsets with more capabilities such as hand tracking, spatial mapping, object recognition, and Large Language Model (LLM) integration. At the same time, device designers have focused on how they can achieve these results while keeping the costs affordable. As a result, powerful AR/VR devices have become much more financially accessible to both consumers and creators.

Yet, even though new capabilities allow HMDs to engage with new use cases, it does not guarantee user adoption. Everyday consumers still seem to be searching for the meaning AR/VR can bring. AR/VR headsets have a hard time gaining mass appeal (Brookes, 2025) Analysts have gone as far as to say that AR/VR headsets are “appealing to only a small segment of video gamers.” (Vanian, 2026).

Technology-centered design is a term interaction design scholars use to scrutinize the above approach. Paul Dourish explains that with technology, meaning making is less about engineering prowess and more about social entanglement and user guided interaction (Dourish, 2004). Bill Buxton urged designers to move from “object-centred to experience-centred”

(Buxton, 2010, p. 10). Jeffrey and Shaowen Barzell say that experience focused design thinking is vital for technology to move into everyday use in life (Bardzell & Bardzell, 2015, Chapter 5).

2.2 A case for contextual design

Hugh Beyer & Karen Holzbolt went as far as to say that technology needs to be designed to the context of the user's specific situation if it is to be meaningful (Beyer & Holtzblatt, 2009). This is the basis of contextual design.

Developing AR/VR device capabilities can facilitate more novel use cases but it won't inherently lead to meaning in users' lives. There are many AR/VR designers that pursue contextual design in AR/VR uses. Through the 80s, 90s, and into today, AR/VR has had application in industry, health, and education (Balalle, 2025; Tan et al., 2022). These are examples of designers making custom AR/VR applications for specific contexts. Yet, it also reinforces the story that AR/VR is mainly for niche problems.

At the same time, contextual design inherits a power imbalance between designer and user. Beyer & Holtzblatt worked to address this power imbalance via contextual inquiry techniques (Beyer & Holtzblatt, 1998, Chapter 3). These techniques are to help designers extract knowledge from the user, a domain expert. But if the user already has the domain knowledge, and technology can enhance their situation – why not simply step aside and let them build AR/VR solutions themselves? The reason lies in AR/VR literacy.

2.3 User literacy and empowerment

In written language, literacy is understood as being able to both *read* and *write.* Another way to see this as being able to understand what you are seeing when looking at written words and then being able to produce words of your own. Literacy can be applied to software as well (Kay, 1989). There is a cyclical nature to gaining literacy. One must be able to read to write but ability to write reveals deeper systems at play, deepening one's ability to read. Casey Reas and Ben Fry explain this and many other nuances in teaching people to become literate in digital mediums (Reas & Fry, 2014, pp. 1–7).

Literacy somewhat relates to the ideas of affordances discussed in the introduction section of this thesis. The author sees the ability to read a material as the ability to recognize its affordances. Being able to write with a material is being able to intentionally manipulate the affordance.

AR/VR literacy is difficult as it is made up of a myriad of both physical and digital materials. The affordances and capabilities in modern HMDs are not clear to the general public (Ipsos, 2023; Katins et al., 2025, p. 173). Even when exposed to AR/VR experiences, the materials are often designed to fade into the background so as to maintain immersion. This makes it harder to see

how the same affordances may be applied in a new way between experiences, obscuring affordances.

For laypeople and newcomers, *writing* in a digital medium requires advanced digital tools. Designing tools for user independence and free expression is a tricky balance of making complex systems approachable while also teaching technical skills. Navigating that balance correctly allows for the digital medium to find new social meaning beyond the limited view of the tool designer (Cuartielles Ruiz, 2018). In the end, the tool allows the author to be spontaneously creative within a digital medium and mold the tech to their individual use context.

The tools that provide authorship play a massive role on who is able to become AR/VR literate. In section 4.2 of this thesis, several AR/VR author tools are explored. Furthermore, this thesis explores how *playful spontaneous creativity* in tools connects to medium literacy.

2.4 What is playful spontaneous creativity?

In interaction design there is something more to spontaneous creativity than just a fast interaction loop. Paul Dourish explains that interactions emerge from situated, improvised action. He argues that meaning in digital artifacts is generated by actions in the moment, more so than predefined system procedures (Dourish, 2004). This thesis understands the user-generated improvisational interactions as framing for *spontaneity*.

Gaver goes a step further and says that systems should be designed to support this type of improvisation and playful exploration (W. W. Gaver, 2002). He holds that affordances in a system are invitations for playful discovery of a technology's use (W. W. Gaver, 1991). Sicart understands play as an embodied way of understanding an object (Sicart, 2016). Reas and Fry sketch with the medium is needed to understand it (Reas & Fry, 2014, p. 2). The author understands this as that, optimally, one should be able to explore AR/VR while immersed in it.

This type of undirected, open, free exploration of a technology is what this thesis considers *playful spontaneity*. Furthermore it sees playful spontaneity being connected to the concept of epistemic play. Epistemic play has its roots in older scholarly work that holds that play is a way of moving through the world and learning about it (Caillois et al., 2001; Huizinga, 1971; Nilsson et al., 2018) When brought to digital mediums this type of play helps us understand complex systems and appropriate them in new ways.(Sicart, 2016, Chapter 8).

Both Gaver and Sicart explain that such play and exploration leads to users adapting the technology to their own needs and uses (W. W. Gaver, 2002; Sicart, 2016, Chapter 8). Such adaptation lets users redefine the meaning of a technology beyond expectations of the tech's designers. This sometimes manifests as appropriation of the technology, meaning that power moves from the designers to the users (Eglash et al., 2004). Hacker culture has its roots in playful appropriation of electronic systems (Levy, 2010, Chapter 1) Seamful design is an approach that intentionally makes openings for appropriation or *seams and scars* (Storni, 2014). In section 2.7 of this thesis, specific examples of this are examined more closely. Appropriation

and adaptation of a technology, and inventing a new user-generated meaning of it, is what this thesis views as *creativity*.

The line of improvised interaction and playful exploration leading to appropriation and new meaning is what this thesis sees as *playful spontaneous creativity*. Later in this thesis, design artifacts that exhibit playful spontaneous creativity are explored for insight as to how it could manifest in AR/VR.

2.5 Sketching, Rapid prototyping, and Epistemic Play

This thesis uses common interaction design methods to examine how to support inviting others to use those same methods in AR/VR. As such, it is important to provide context as to how certain design methods align with the pursuit of *playful spontaneous creativity*.

Sketching sits at the base of most art, design, and generally creative pursuits. While many imagine paper and pencil as how one sketches, Bill Buxton argues that the term applies to a much broader range of practices. They provide criteria for sketching as a mode of working through solutions in a flexible, non-committal way It allows for spontaneous redirection and repurposing of ideation. (Buxton, 2010). One could say that sketching is quite literally playing with an idea.

On the other hand, prototyping is a practice about asking questions (IDEO, 2015). As prototypes inform designers, they make changes to their design. Prototyping is in pursuit of a use case goal. A productivity mindset directs exploration, inquisition patterns, and analysis (Houde & Hill, 1997).

In rapid prototyping this happens quickly, leading to spontaneous redirection of the project. Some design researchers see it as a blurry line between sketching and prototyping (Houde & Hill, 1997; Moussette, 2012). Buxton holds that sketching is very different from practices like rapid prototyping. He explains that the key difference lies within the “... how, when, where, and why”(Buxton, 2010, p. 383). This thesis sees play as a key difference in the *how* and the *why* that discerns sketching from prototyping. Like prototyping, play is exploratory and inquisitive, but it has a more ambiguous nature with less stress on real world consequences (W. W. Gaver, 2002; Sicart, 2016).

This thesis sees sketching more related to a form of playful inquisition and exploration. It sits early in the process and allows designers to tangle with what problems to solve in the first place. In this way it plays a major role in “designing the right thing” instead of “designing the thing right” (Frayling, 1993).

As mentioned in section 2.3, being able to sketch within also leads to deeper literacy of the medium. However designers must make tools that allow for it (Cuartielles Ruiz, 2018; Reas, 2007).

AR/VR is an immersive medium. Its authorship tools are a mixed bag of immersion and non-immersive design processes. This thesis further into how immersive sketching is currently presented and could be improved.

2.6 Seamful Design

Seamful design holds that users can weave technology into their unique context if allowed to see openings (Chalmers & Galani, 2004; Storni, 2014). A seam in your shirt shows where you can remove a stitch and alter the clothes. The seam also gives evidence on how the fabric was put together to form a shirt. In digital artifacts, that seam or scar must be actively built by the designer. This comes back to Gaver's statements about affordances being invitations in digital artifacts (W. W. Gaver, 1991). For AR/VR, different affordances in the medium need to be revealed as alterable through an intentionally made seam or scar in programming tools. The key to it being a seam is that the seam is visible (Storni, 2014). Both play and spontaneity rely on the designer to know what is available. Therefore tools that technically allow for alteration, but do not feature visible invitations to do so, do a disservice to creative users – it fails to show them what they *could* do. This disproportionately affects laypeople as their lack of experience makes affordances even more invisible. This thesis sees making obvious seams as a teaching tool and an invitation for users to sketch and play with an affordance of the medium.

2.6.1 Seamful design and constructionism

Constructionist learning offers an understanding of how people can learn through the act of building (Papert, 1987). This concept has been studied in instructional technology (Parmaxi & Zaphiris, 2014). Research shows how this approach has application in helping people of all ages approach technically challenging skills through building and co-creation (Roque & Tamashiro, 2022). Seamful design allows for users to learn about how things work by opening, rebuilding or repurposing it (Storni, 2014).

Much inspiration can be gained from examples in the tech built for teaching. Arduino is a microcontroller made for learning, Fig 3. Processing is a coding environment to teach coding to designers. Both aim to make technology approachable enough to learn (Cuartielles Ruiz, 2018; Reas & Fry, 2014). There is a similar empowerment energy in the larger maker movement which looks to bring creative power to individuals (Papavlasopoulou et al., 2017).

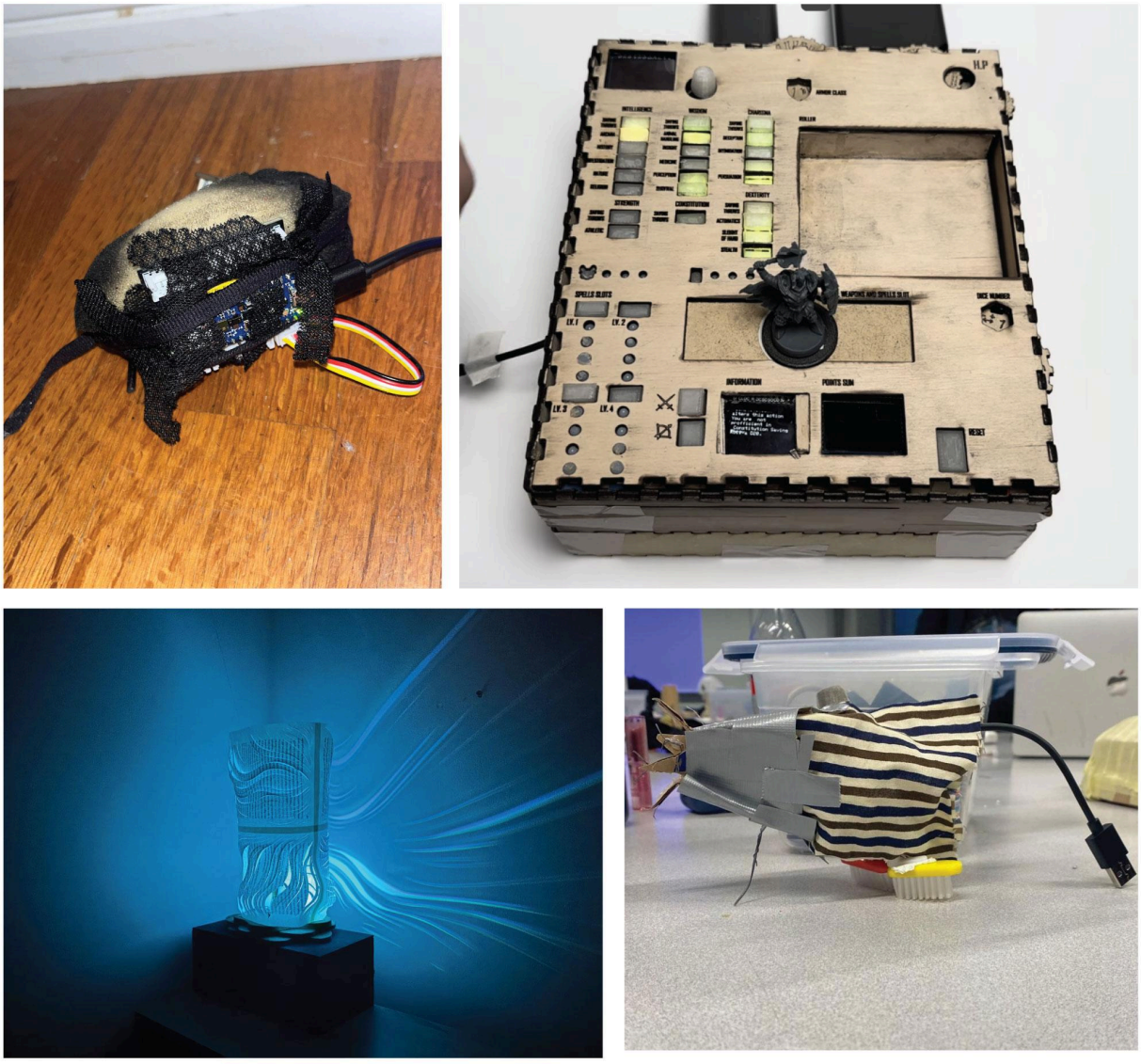


Fig 3: Arduino based student projects for learning electronics in the interaction design program at Malmö University. Images provided by students with permission for use in this thesis.

2.7 Related Arts

2.7.1 Seams at play in digital mediums

Appropriation of digital artifacts is already seen in mediums close to AR/VR such as video games. Sandbox video games, such as Minecraft, are appropriated for education (Alawajee & Delafield-Butt, 2021). This is no surprise as it is built to be appropriated. The code is open to be hacked and modded on purpose. There is even a store front hosted by the owners to platform these user made versions (Filby, 2024). The game's designer also intentionally blends game

world modification into the gameplay loop, making it “...something between a game and a tool.”(“Markus Persson,” 2012)

Other games feature attached tools and modes to help you build with them. Warcraft 3 Map Editor, a tool attached to the popular video game Warcraft 3, enabled players to invent the multiplayer online battle arena genre that dominated the video game market for a time (Atkinson, 2018). Fortnite and Roblox are two more games that have turned players-as-creators into part of their business model (Shepherd, 2025).

Part of why this works can be seen through the lens of epistemic play discussed in section 2.4. While playing the game, they learn how the complex systems react. Tools built into the system act as seams that let the players turn knowledge gained from play into novel uses of the systems.

This can be seen very clearly in Zelda: Tears of the Kingdom (Nintendo Co., Ltd., 2023). It features a fusing system, which inspired aspects of this project’s final prototype. In Zelda Players can fuse objects with specific properties to create novel tools and overcome game challenges. The objects themselves are visually coded as real world objects, a wheel that turns for example. These basic parts are used to complete puzzle-like challenges in the game.



Fig 4: A player made perpetual motion machine that breaks the physicist engine of Zelda: Tears of the Kingdom. Image used under fair use for scholarly purposes. Original found in Screenrant article by Lee D'Amato, https://screenrant.com/zelda-tears-kingdom-vehicle-builds-broken-physics-totk/

Players have pushed this system to build things beyond solving simple puzzles, see Fig 4. By stacking different basic pieces together, players make novel game-breaking creations. The first obvious use case telegraphed an affordance of the individual part. The system made seams through the fusing function, making the user create their own objects as solutions to a challenge. As the users built and problem solved they became more literate of the system as a whole which allowed them to appropriate it for novel use cases.

The final prototype of this thesis, Alchemic Reality, explores how a sandbox AR experience can leverage this same approach to seams for AR/VR literacy.

2.7.2 Current AR/VR focused tools

This research project also engages in a review of several current AR/VR authorship tools. For creators and builders, knowledge lies within artifacts. That knowledge is extracted through engagement with the artifact (Frayling, 1993). These experiences serve as a repertoire that influences design decisions (Schön, 2000). As such, this thesis looks at related arts as a type of literature to inspire tool development.

The specific tools this research images with are Unity (Unity Technologies, 2025) with the Meta SDK (Meta Platforms ,Inc, 2025), Shapes XR (Shapes Corp, 2025), FigmenXR (Overlay, 2024), Recroom (AgainstGravity, 2025), and Google's VibecodingXR (Google LLC, 2025). In section 4.2 of this thesis these tools were looked at more closely.

3. Methods: Research Design & Structure

3.1 Methodology

This research project follows a Research through Design (RtD) methodology. RtD comes from the idea that craft and building are a form of knowledge building and exploration (Frayling, 1993; Schön, 2000). A formalized format for HCI as outlined by Zimmer et al.(Zimmerman et al., 2007). The resulting contributions are an AR/VR authorship tool prototype, Alchemic Reality, and this thesis which documents process and relevant scholarly insights.

3.2 Specific Methods

Many methods used in the project are explained in section 2, in the context of how tech designers benefit from the methods. The following expounds how these methods were also practiced within this thesis project and introduces other methods used.

3.2.1 Expert interviews

Interviewing a wide range of experts play a crucial role in helping understand the wider application of AR/VR as a medium quickly (IDEO, 2015). Six professional practice AR/VR experts in or near southern Sweden were interviewed. Their experience ranged through immersive art, mental health exercises, interactive education, immersive sport streaming, immersive documentary, and AR platform development. One expert was a builder that used Minecraft as their crafting tool to design educational experiences. This expert was not interviewed for their views on AR/VR rather as an expert on appropriation of a digital medium.

Time did not allow for a true shadowing style the expert interviews found in contextual inquiry (Beyer & Holtzblatt, 2009). Instead the researcher engaged in these interviews as a student who wanted a brief overview of how the expert approached the medium. As a builder themselves, the researcher inquired as someone who was looking to inform their own processes rather than just as an outside observer. Though no sensitive personal information was collected, any mention of specific projects they have made has been intentionally left out to provide anonymity.

3.2.2 Audience Surveys

My former students at Malmo University were my target audience. They had recently engaged with learning VR/AR production for the first time. They were surveyed to inquire into the recent learning process with the tools they engaged with. During the class many of these students had already informally talked about challenges they encountered. However a more concrete survey gathered what insights remained after letting the dust settle.

Surveys were done online and answers were anonymized. Answers were manually analyzed for patterns between participants. No AI summary tools were used for analysis and synthesis. As

per GDPR requirements their names have been anonymized. Similarly their answers have been mainly summarized instead of quoted to avoid mention of their specific projects.

3.2.3 Material Exploration and Critical Making

There is an important distinction between the materials being explored in phases 1 and 2. In phase 1 the materials in the medium of AR/VR tools are being explored via interviews with experts and surveys of recent students. In phase 2 the affordances does the tool allow the user to manipulate and what way does it approach those affordances.

Phase 2 of this project engages in a material exploration method closely related to critical making. It uses building with tools while looking through a material lens, built in phase 1, as a form of material analysis (Ratto, 2011). This project however does not engage with co building, which is often found in critical making. Instead it relies on expert interviews to establish the material lens of AR/VR and then use micro-critical-making sessions to examine the affordances within tools around the material.

In addition to testing the tools myself, I used desktop research of user reviews to identify common frictions found within the tool. It was impossible to tell what problems occurred due to the experience level of each reviewer. So user reviews were used to see what problems seemed to repeatedly appear across the audience.

3.2.4 Sketching

Beyond classic forms of sketching with pen and paper, theatrical forms of sketching were used (Buxton, 2010, p. 267). All hand gestures were first explored by performing them while imagining what each action might do. Steamed glass panels of a shower door were used to sketch how floating interfaces might react to specific hand motions. This helped imagine things appearing and then being quickly wiped away.

Part of the visual coding system was drawn on paper cut outs and connected with ribbon. These were used to sketch and eventually prototype different connection methods.The researcher considered this as sketching more than prototyping as it was mainly meant to playfully explore how the parts of the system could connect. It was quickly made and not shown to others. It had notes attached to help think out where chunks of code would need to sit and what information it would need to pass on to other parts of the system.

3.2.5 Prototyping

The thesis does not see iterative prototyping as a version of sketching as discussed in section 2.3. However the lens of making a prototype for a specific purpose leans on the understanding given by Houde & Hill (Houde & Hill, 1997). Questions asked to test users of the final prototype focus heavily on role and look & feel. Efforts at efficient implementation on the coding level are not a priority. The final prototype acts as a means to compile everything learned into an artifact

that personifies the knowledge explored and produced from the project (Bang & Eriksen, 2019; W. Gaver, 2012; Zimmerman et al., 2007)

3.2.6 Playtest sessions

Early and iterative playtesting of a game are key in locking down the key experience of its playful experience (Salen & Zimmerman, 2004, Chapter 2). Informal playtests happened early and frequently but formal playtests were also organized near the end.

The users were told to engage in a playful manner. This encourages users to explore the boundaries of what the tool is capable of and leads to seeing emergent behavior (Flanagan, 2009). Users were not only asked to try out specific mechanics. They were given periods of un-mediated play and eventually were asked to try and break the game.

Informal interviews during and after the interviews were recorded and will be deleted by time of publishing of this thesis. Names were intentionally left out of recordings and any quoted response to questions are attributed an anonymous name to protect user identity.

3.2.7 LLM Usage

This project engaged with large language models in a few ways. Firstly it was used as a consultant to understand concepts of the code library added through the Meta SDK. Secondly, LLMs were used to look for scholarly papers around key concepts. These sources were then read by the researcher to gain conceptual knowledge that appeared in this thesis. No text in this thesis nor code in the prototype was generated by an LLM.

VibecodeXR is a LLM based coding tool for AR/VR creation. It was used while being reviewed as part of of phase 2 of this research and talked about in several other places in this thesis.

3.2.8 Sensitive data collection statement

No sensitive data was collected during this thesis. All names and identifier information has been obscured as far as is reasonable. Any answers to surveys will be deleted upon completion of this thesis. All recordings that include peoples likeness will also be deleted. Any images that may contain testers in this thesis are used with permission by the tester.

4. Design Process

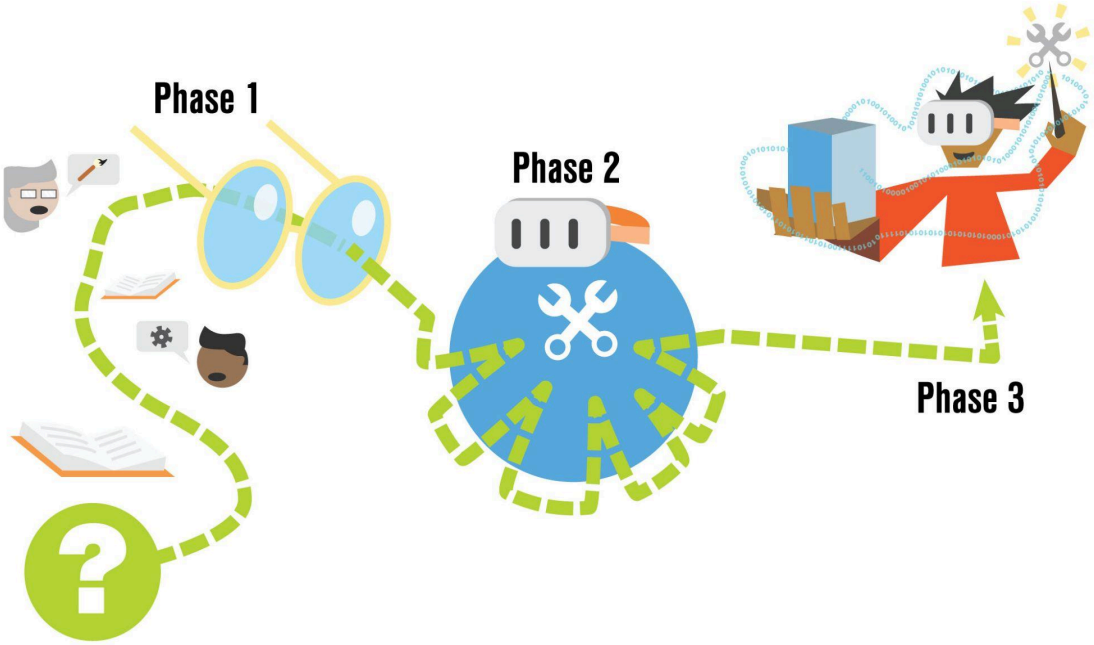


Fig 5: Research phase 1 used methods to establish a lens. Phase 2 used methods to explore the current toolscape using the lens. Phase 3 turns findings into a playful AR/VR tool.

The project was split into three phases loosely based on the programmatic structure outlined by Anne Louise Bang and Mette Agger Eriksen (Bang & Eriksen, 2019, p. 4.8-4.9). Phase 1 was largely exploratory and established a research lens. Phase 2 stabilized the research around the problem area. Phase 3 finalized and proofed findings. The phases had soft edges, meaning the process of each bled into each other to some degree.This was especially true between phases two and three, as the research intentionally jumped back and forth between them.
Goals and milestones for individual phases are outlined below. A brief description of what key findings came from each phase. These findings are discussed at further length in section 5.

4.1 Phase 1

This phase was fairly exploratory in trying to establish a lens of AR/VR on a material level. It was focused on research question one, trying to block out what materials commonly exist within AR/VR. The methods leveraged were: Literature review, expert practitioner interviews, and surveys of recent AR/VR beginner designers.

Marika Hedemyer's understanding of the “strands” of mixed reality (MR) offered a starting point (Hedemyr, 2023, pp. 173–178). The researcher’s personal experience building brought Hedemyr’s lens closer to applying to the use case of building for a Meta Quest 3.

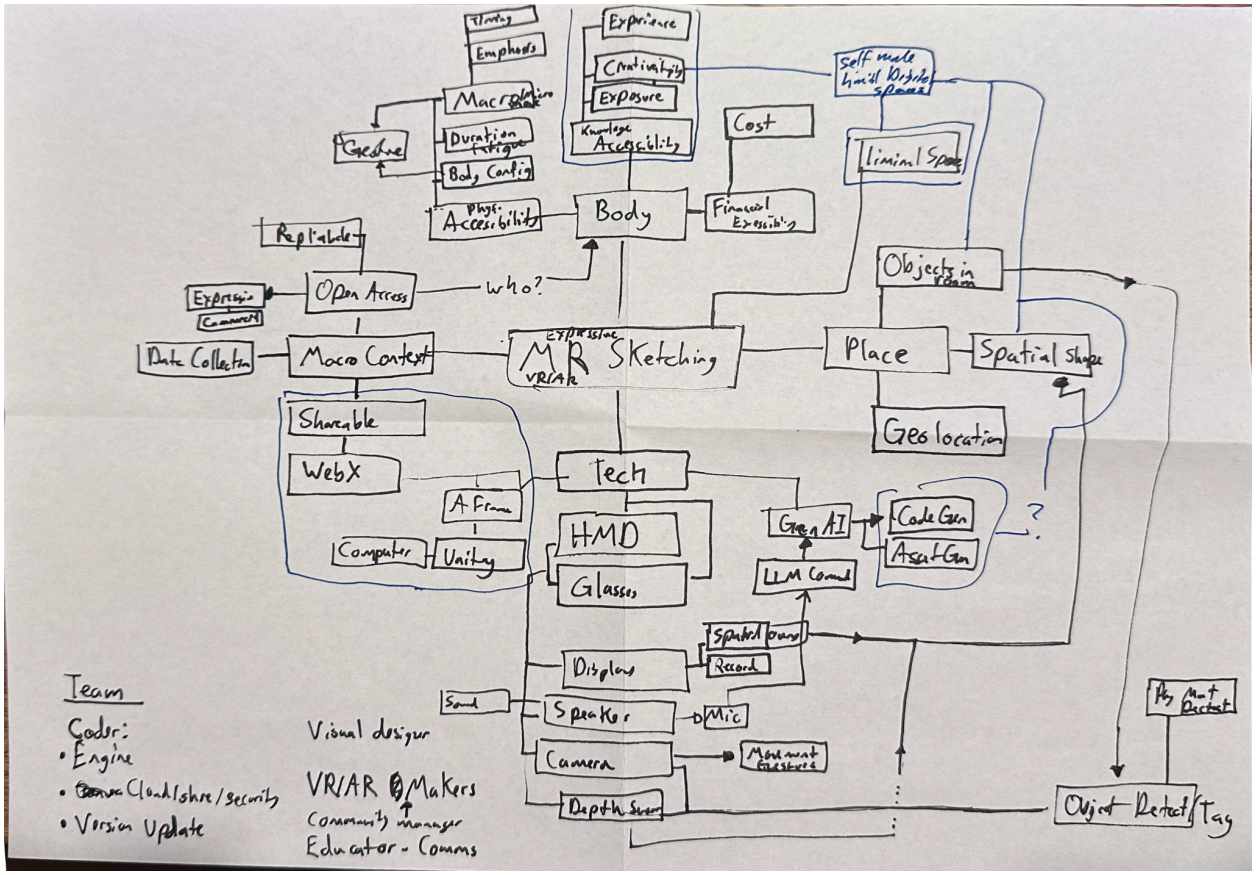


Fig 6: A sketched mapping of materials and how they relate made by the researcher during a workshop with Marika Hedmyr.

Insight from expert interviews broadened the lens of materials and affordances available in AR/VR. They also gave insight into design decisions that can guide the user to intended interactions.

Target audience surveys asked about what materials they saw as most exciting to engage with. These gave insight into the reality of how personal technical skill level and tools governed which materials they were able to engage with.

The result was a suggested mapping of the AR/VR materials that comes from a blend of personal experience, scholarly insight, professional practice, and the budding hopes of new designers. Fig 7 shows an overview of the materials identified. Appendix B gives a more detailed explanation of specific identified materials.

Next Page Fig 7: A list of the identified materials within AR/VR organized into different overarching categories. Colorful squares mark who mentioned each material. Wrenches mark where digital tools were seen as able to intervene.

Augemented Reality Materials

User Body

- Gestures
- Mobility
- Memory
- Creativity
- Group size
- Tacit Knowledge
- Stamina

Site/Environment

Physical Reality

- Spatial Shape
- GeoLocation
- Objects Present

Virtual Reality

- Spaces digital twin
- Tele Location
- Digital Proximity

Tech/Media

- Display
- Camera
- Depth Sensor
- Controllers
- Gesture Detection
- Computer Vision
- CPU/GPU
- Speaker
- Mic
- LLM interfacing
- Streamed Video

Storytelling

- Graphics (3D objects)
- Text
- Audio
- Haptics
- Co-creation
- User Interface

Over time

- Code Logic
- Animation
- Time tracker

Virtual World update

- Reactivity
- Passive update

Physical World update

- Reactivity

Passive update

- Captured User Data

My TP1

Jeannette Ginslov (VR, Art, Therapy/Medicine)

Bram Wultepute (Virtual world designer- Minecraft, Interactive Education)

Jakob Skote (Mobile AR, AR Art sharing platform)

Sandeep (VR/AR Events, Immersive sprots broadcasting)

Students signal desire or attempted

Tool intervention

4.2 Phase 2

The goal of this phase was to mature and stabilize the research. Desktop research and information from expert interviews procured a list of potential tools to examine, mentioned in section 2.7.2. A lens was made by converting the list of materials and affordances in Fig 7 into a questionnaire. The examples of these questions can be seen in Fig 8 and 9. A full list of the questions are found in Appendix C.

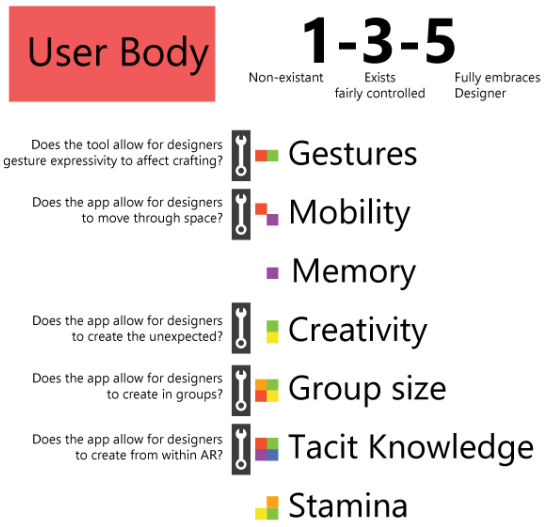


Fig 8 : Focuses on how users can use their bodies



Fig 9: Focused on authorship of interactive

Analysis of a tool was focused on which AR/VR affordances it exposes and how easy they were to manipulate. A 1-5 scale shown in Fig 10 was established to gauge if access to affordance existed through the tool but also if it was quickly exposed to the user. Notes were also taken as to how playful or sketchable the affordance was in the tool. Time did not allow for this gauge to be an empirical analysis, rather it was based on how the experience seemed in the short review period.

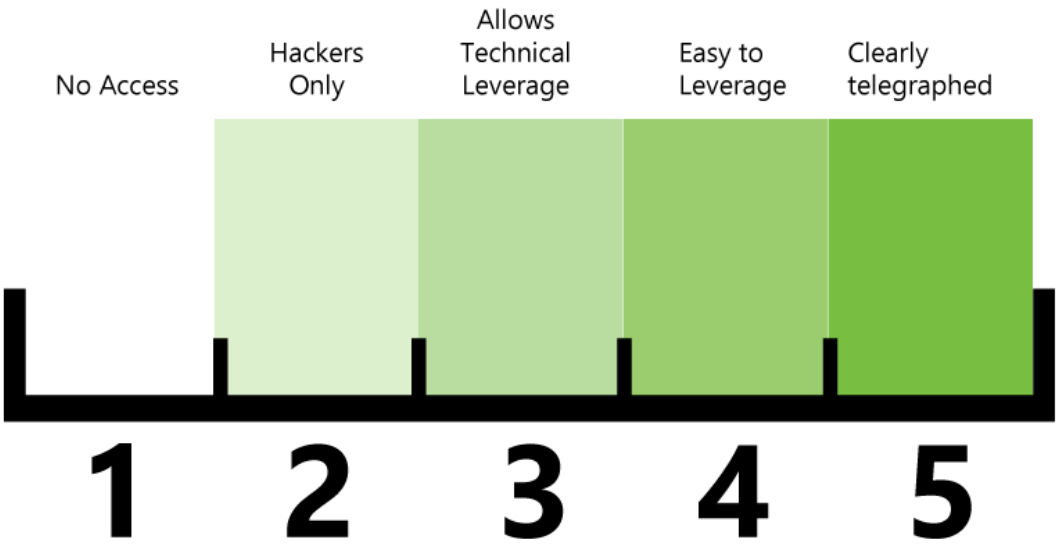


Fig 10 A scale that gauged if an affordance was available and then if it was also approachable within the first few days of learning the tool. Image made by the researcher.

Desktop research first examined a tool's capabilities via website documentation and user reviews. Unity and ShapesXR were familiar enough to review based on the researcher's past experience. For these the researcher also attempted to frame review from where students had run into friction.

A few other tools were further explored by building with them for a few days. They were: Rec Room, VibecodeXR, OpenBrush, and FigmenXR. For these a tutorial period was spent to learn the basics of the tool, followed by a short period of building with it. The condensed learning and building schedule was used to help frame what was approachable within the beginnings of learning a tool. Deadlines were also used to ensure the researcher would be able to move through multiple tools.

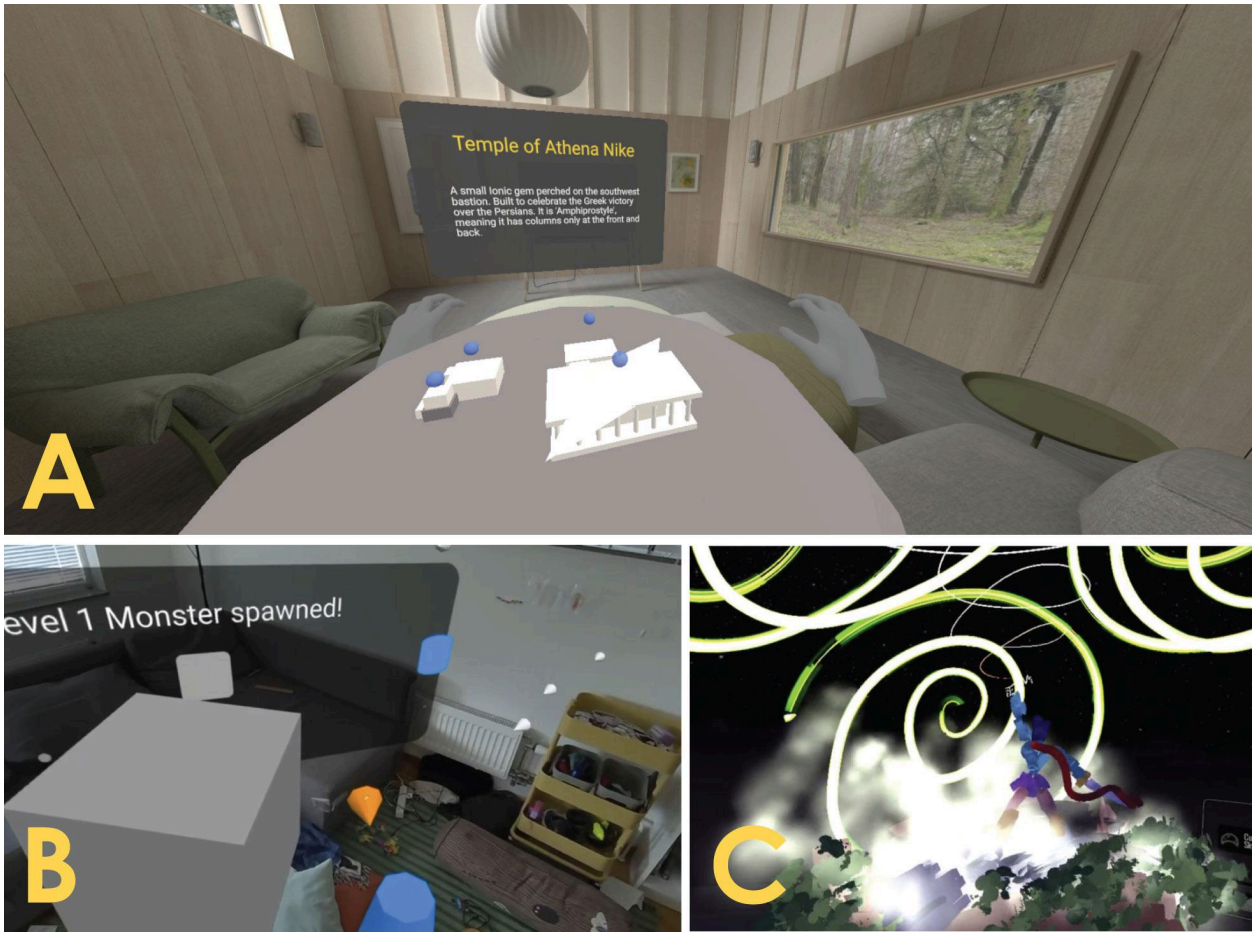


Fig 11: Several smaller AR/VR creations made while reviewing tools. A: Interactive history viewer of the acropolis, via VibecodeXR. B: spatial tower defence game via VibecodeXR. C: OpenBrush painting with special visual effects. Not pictured a space blaster in Rec Room and a moving airship in Figmen XR. Screen shots provided by the researcher.

At the conclusion of reviewing the tools, the researcher analyzed notes in the questionnaire for patterns in how individual affordances were exposed across tools. This provided an informal mapping of where gaps between tools might exist for new students or laypeople.

The questionnaire analysis was cross referenced with what affordances students found most interesting during the survey discussed in section 4.1. Ideally a comprehensive tool that gives seamful access to all affordances would be built. For the scope and time frame of this thesis project, only the following gaps were further explored.

Affordances in computer logic around interactions in AR/VR was identified as both hard to sketch and of interest to the students. It seemed that there was very little ability to modify computer logic through an immersive tool.

There was a gap in interacting with aspects of the physical world, such as the gesture based input and spatial models of the environment. Many of the students had shared this as an area of interest but expressed it was out of reach for their projects.

4.3 Phase 3 Final Prototype

This phase was focused on closure and finalization. An initial period was spent creating a final design artifact mixing findings from phases 1 and 2. This prototype was geared on providing students and laypeople a gateway into exploring logic building, spatial interaction, and body based interactions in AR. This prototype was mainly focused on exploring the role of such a tool – what effects does it have on AR/VR literacy? The prototype also had a fair amount of look and feel as to gauge how easy it was to interact with the system.

The researchers tool set of choice was the Unity game engine with the Meta SDK. This was based on the researcher's experience level with the tool. The SDK allowed for access to all the capabilities of the Meta Quest 3, the target publishing device. The SDK also provided standardized building blocks that give basic functionality connected AR/VR experiences. These building blocks often had to be modified with custom code to make them work as desired.

As mentioned in Section 3.1, RtD is an attempt at gaining knowledge through crafting and building. As such it was important that the researcher participates in the building as much as time and project scope permit. In this prototype, LLMs, Co-Pilot and Gemini, were used as code base consultants. Vibe coding was not used for the final prototype outside of a few lines of copied code from examples. This set up enabled investigation of how to make the prototype work while letting the researcher remain close enough to the building to gain insights.

4.3.1 Sketching and planning

The first week was spent sketching and planning. Paper and pen sketching provided preliminary ideas. A gesture based building system was imagined as a hands-on way to inform the user about subtle affordances in gesture detection. Eventually more theatrical forms of sketching were needed to explore the gesture based system. To keep interactions with the system playful, these gestures had to be felt out first and for most.

An initial version of the prototype was made over the course of two weeks. Development was split into two parts, spatial sketching with objects and sketching with interaction.

4.3.3 Spatial Objects and Sketching with Gestures

Immediately after adjusting the basic settings of Unity and the Meta SDK to be testable on the headset, spatial visualization functionality was added. Tests of the system were always run displaying both the passthrough information of the camera along with a visualization of the spatial data stored in the system, see Fig 13.

Basic manipulation of primitive 3-dimensional(3D) objects using the players hands was added first, utilizing Meta's out of the box building blocks. Advanced custom gesture controls were added iteratively after rounds of testing by the researcher and close friends. Fig 11 shows iconography used to telegraph different connected commands.

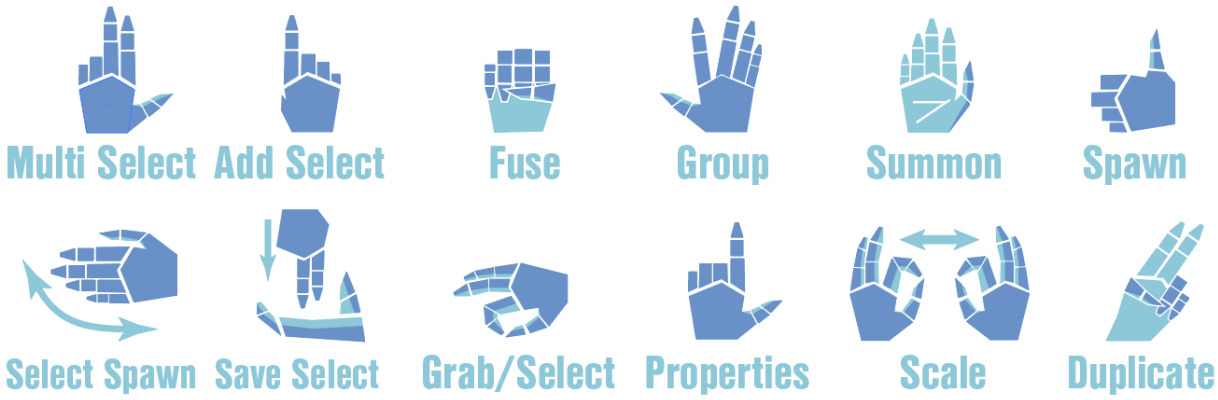


Fig 11: this diagram of the system lived in an Adobe Illustrator document and was regularly updated as a way to track the current state of the implemented system. Image provided by the researcher.

The left and right hands are seen as separate inputs. Not only do finger positions dictate the act but also the facing of the hand. For example, inputs require the palm to be toward the face. Stacking requirements make the system more finicky in detecting whether something is correct, but also in making sure the user is intentionally performing a command.

Some commands require both hands to perform. These commands start with holding a specific pose with one hand. The other hand is then given multiple commands it is able to do. This allows a grouping of similar commands together. For example, any commands that deal with multiple objects at once are grouped within the act of holding the left hand in the *multiselect* pose shown in Fig 11. Holding the pose allows the right hand to either touch multiple objects adding them to the selection. They can also make a fist palm toward the user's face to fuse multiple selected objects into a new object. Releasing the pose sends the system back into single selection commands. This system can be viewed as a state machine.

Properties like shape, color, and the object's reactions to the physics engine are all editable in the Unity editor. These properties were routed into options on a more traditional, 2-dimensional(2D), menu with clickable options in Fig 12.

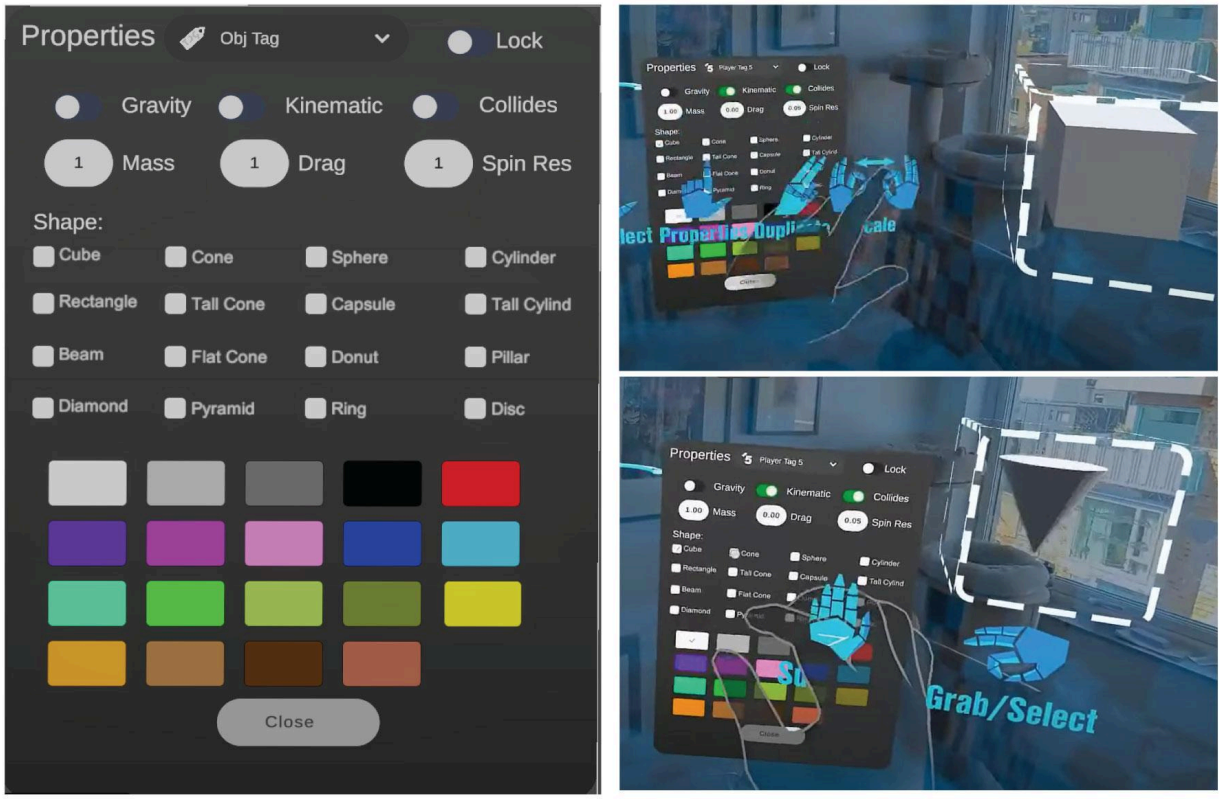


Fig 12: the floating menu used to edit a selected object's reactions to physics, gravity, ability to be moved by other objects, color, and shape. Users can touch options with a finger or pinch their fingers while aiming at the option to select it. Screenshots by the researcher

Informal tests with family and friends played a role early on in testing basic viability of the gesture based building with shapes, see Fig 13.

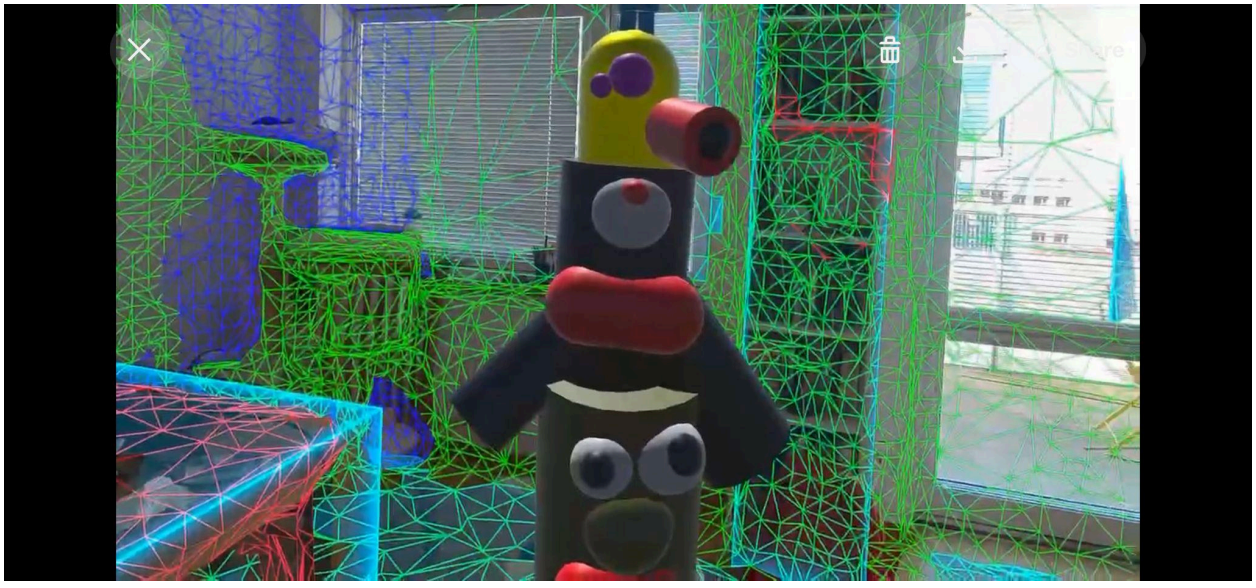


Fig 13: screen shot of a monster created by a youth user testing the gesture building system. Web-like texture shows how the device views the spatial shape of the room. Screen shot by the researcher.

These early testers would often suggest what types of functions they wished they had like, grouping objects, duplicating objects, and deleting them. They also gave insight into the types of shapes they would enjoy building with.

A wizard-like motif emerged throughout the process. This came from feedback during early testing, with users stating, “I feel like a wizard!”. Later references to wizards, magic, and alchemy from pop culture inspired some of the structure of systems within the tool itself. Visuals style and naming of the prototype came from this inspiration.

The many insights and lessons learned from building the gesture based system along with evaluations of its effectiveness are discussed further in section 5.2.

4.3.2 Logic crafting system

The AR/VR materials chart from section 4.1 was referenced as a start to what types of logic could be added to sketch with. The affordances listed under the *Over Time* category seen back in Fig 7 were turned into potential functions in code that connect to that affordance, example below in Fig 14. Along with function types, symbology was imagined for conveying the function. Work then moved from conceptualizing the system into making something that could be interacted with.

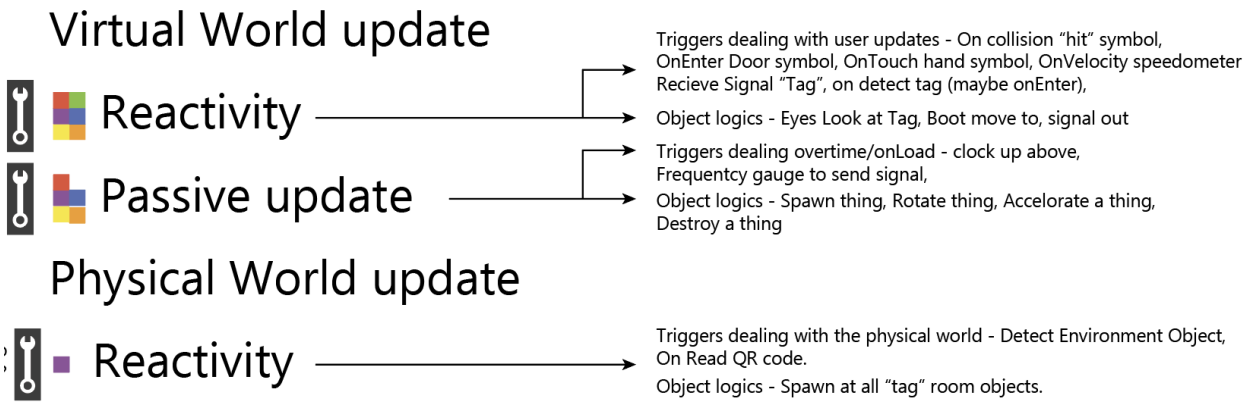


Fig 14: Screen shot of the researcher’s notes. Many standard types of functions exist that relate to specific affordances. They were used for imagining what logic would be needed to give access to the affordance.

In Alchemic Reality, a logic building block system materializes some of these common compute functions into virtual objects to be manipulated with the hands. The system attempts to create a visual language that can act as a seam that signifies where opportunities for logic systems exist in AR/VR.

The system functions much like other visual node based programing systems. Different blocks have slots open for input and can send an output signal to other nodes. Inspiration came from

ActionScript 2.0 (Adobe Systems Incorporated, 2003) featured in Flash (Adobe Systems Incorporated, 2002) some years ago. Flash used object oriented programming in a very visual way. Code blocks were attached to specific graphic objects, called symbols. The symbols could send messages to code blocks attached to other symbols, which could then pass it on again, creating chain reactions.

Symbols could be nested within each other to create a complex network of logic attached to layers of graphics. Likewise, Alchemic Reality attaches visual logic blocks to objects in the scene. It allows the users to fuse logic blocks with visual objects and then connect that logic to other objects if desired.

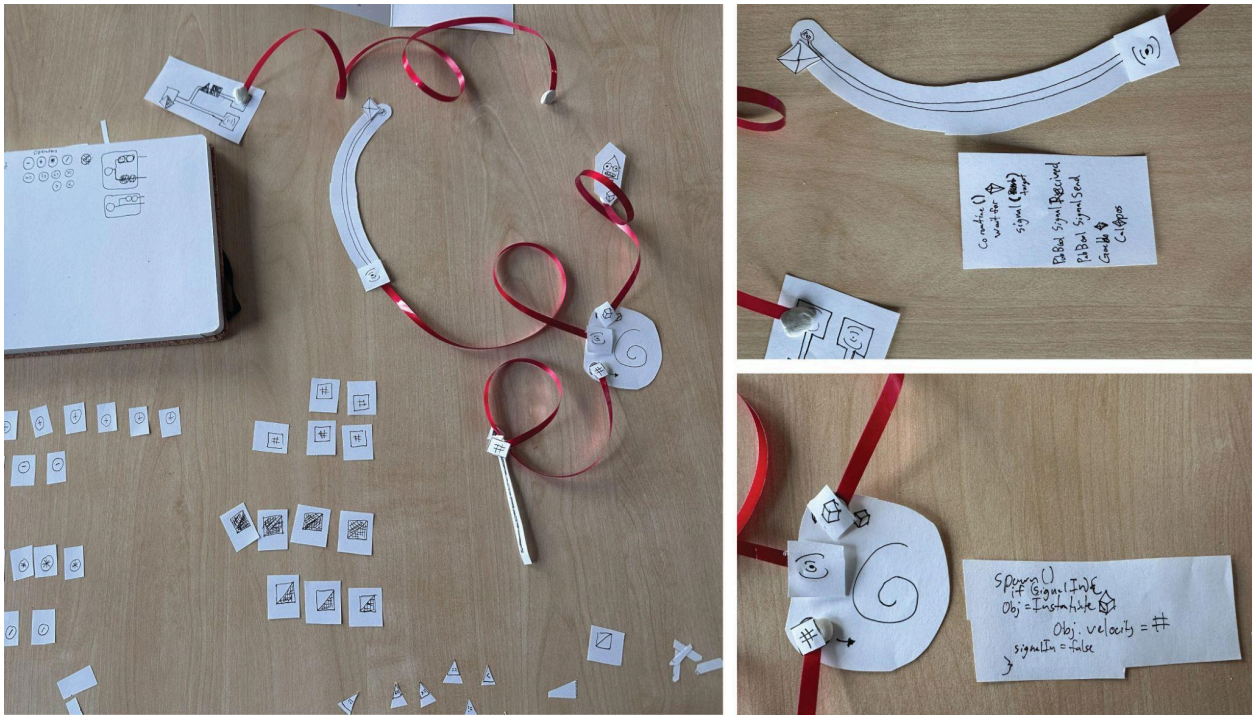


Fig 15: Visual logic chaining system was sketched using ribbon to represent the stings of logic made by a player. Photos taken by the researcher.

The system had to be sketched out and played with, see Fig 15, to gauge how the player might actually string logic blocks together using their hands. Notes on smaller pieces of paper showed what code might fire off within each block.

It was around this time that the visual motif of transmutation circles was integrated. Design specific decisions behind this are discussed further in section 5.3. In short a connection was seen in a basic computational model, input> compute > output, to pop culture framings of transmutation circles, Fig 16. They offered a visual vehicle that matched the wizardy feelings expressed by early testers of the experience.

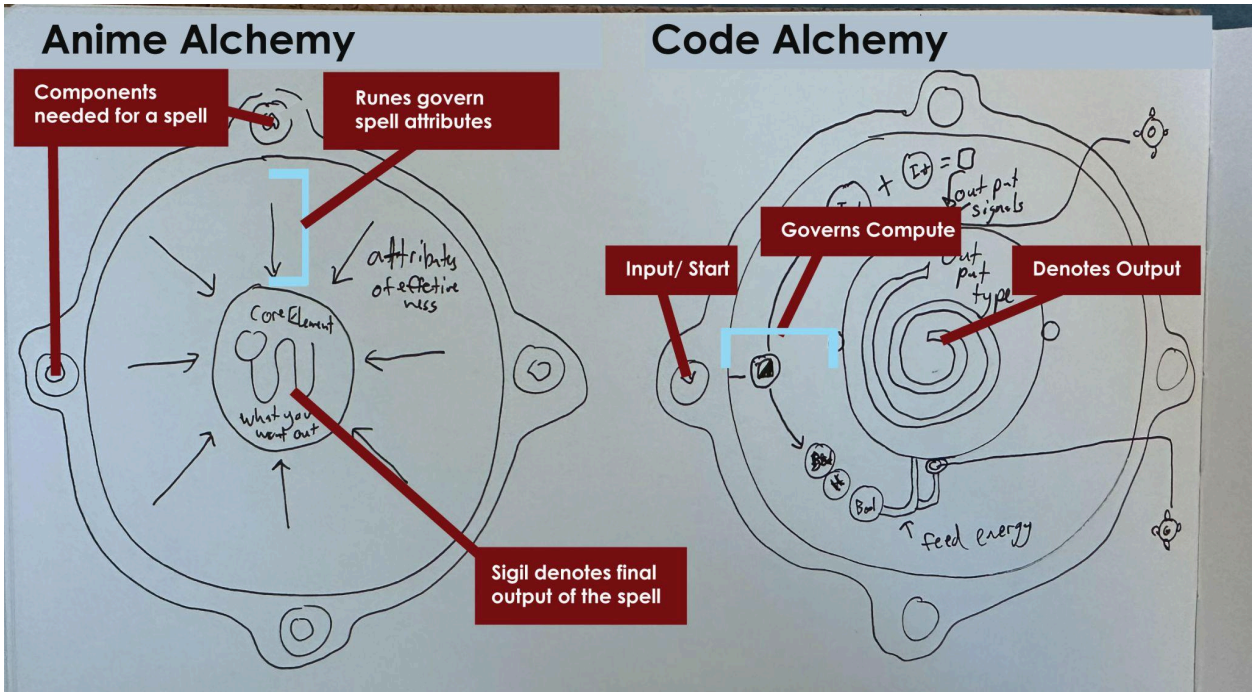


Fig 16: Annotated diagrams from researcher's sketchbook. Left: how transmutation in Full Metal Alchemist (Arakawa, 2001) and magic circles from Witch Hat Atelier (Shirahama, 2016) work as understood by the researcher. Right: How it could represent a computational model.

In Alchemic Reality, each logic block, called *alchemic nodes* hereafter, runs on the same internal system of start on a signal received (input), compute (denoted by symbols on the alchemic node), and an area that denotes or acts as an origin for output. Some alchemic nodes send a signal out when they complete their function allowing for a chain reaction effect.

To ensure chain reactions could be made between all signals, all basic variables in the functions are set to be a float, a type of variable in C# that holds decimal numbers. Unity uses floats to position objects in virtual space. Making all output as a float allows position to be altered novel connections by the player. Floats can be used to track true or false statements, whether it is 0 or not, for logic gates. It can also be rounded to the nearest integer for lists and other internal functions. Tools to adjust floats are represented by a crystal and can be seen in Fig 17.

These implementation choices are not about processing efficiency. They are concerned with system flexibility. When flexibility is the goal, the system can withstand the user combining functions in new ways in a playful manner. Section 5.3 discusses how these choices relate to the project goal and evaluates their effectiveness.

The alchemic nodes and pieces that exist in the final prototype are diagramed in Fig 17-22. Many of them are easier to understand watching them in action via the link in appendix A.

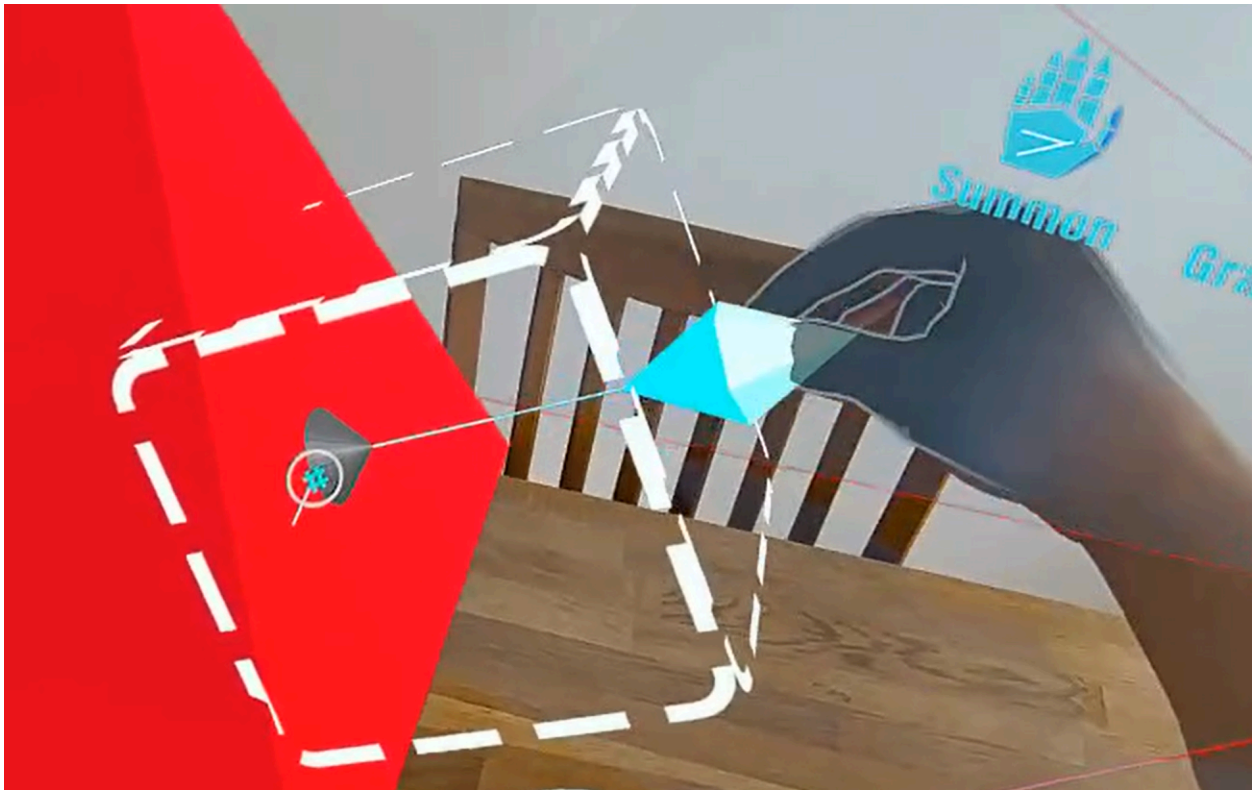


Fig 17: Float adjust tool that sends a decimal number as a signal out. Distance from the base to the crystal governs float amount. Screen shot of researchers' own test session.

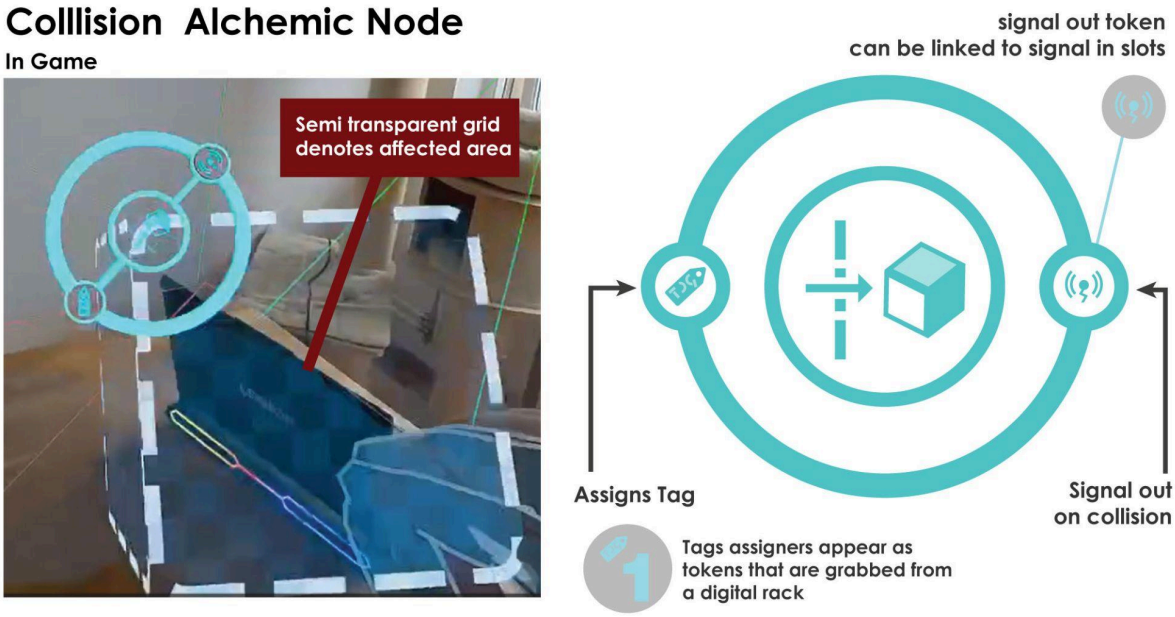


Fig 18: Collision area detects when objects enter its bounds, it checks if the objects tag matches a tag assigned by the player. If they match, it sends a signal out. Screen shot and diagram by researcher.

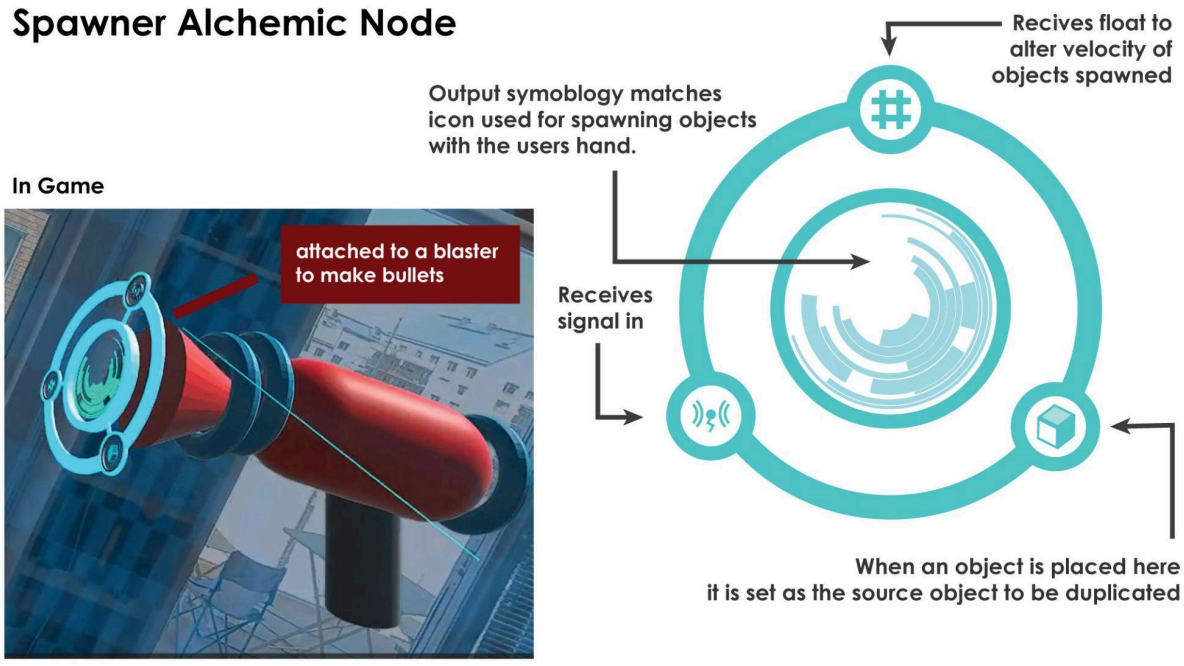


Fig 19: Summon circle duplicates a source object whenever it receives a signal in. This was the first node made as it was just a translation of the logic that let users summon objects for building. Screen shot and diagram by researcher.

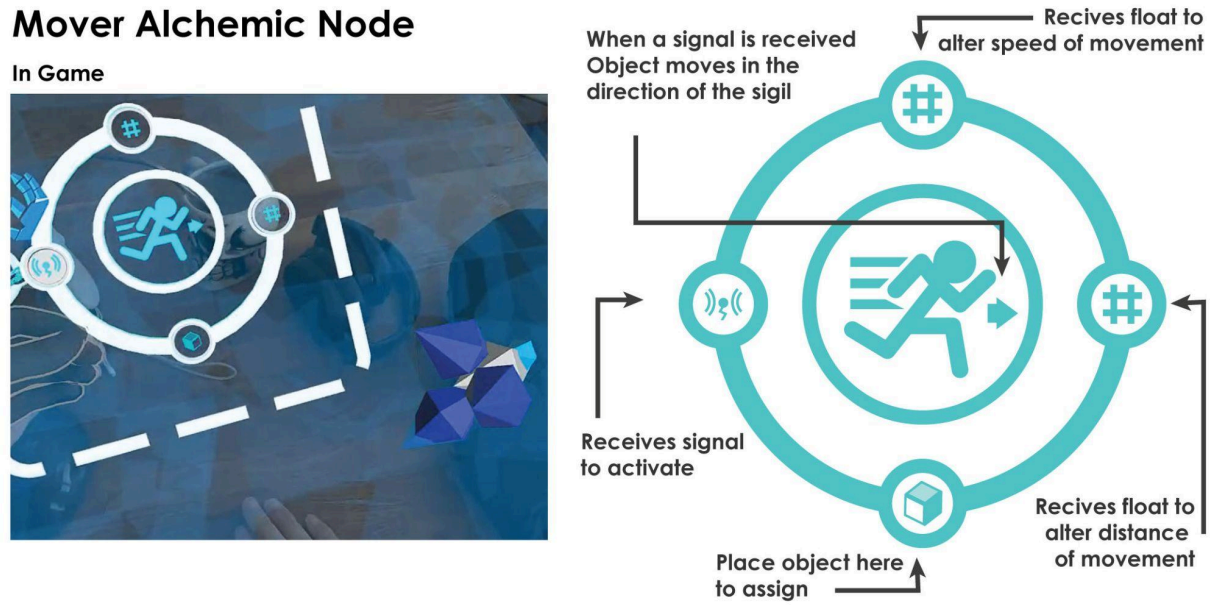


Fig 20: if this Mover node receives a signal it moves assigned objects in the direction of the arrow shown on the central sigil. Screenshot and diagram by researcher.

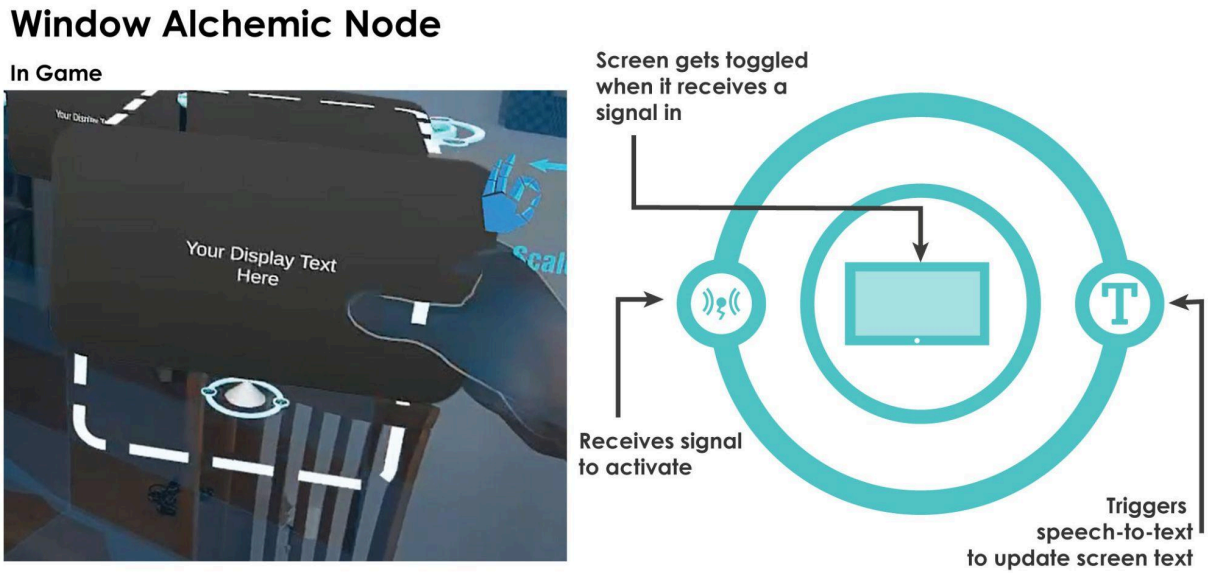


Fig 21: window node was only partially functional at time of testing. Signal in would toggle the screen on or off. Screen shot and diagram by researcher.

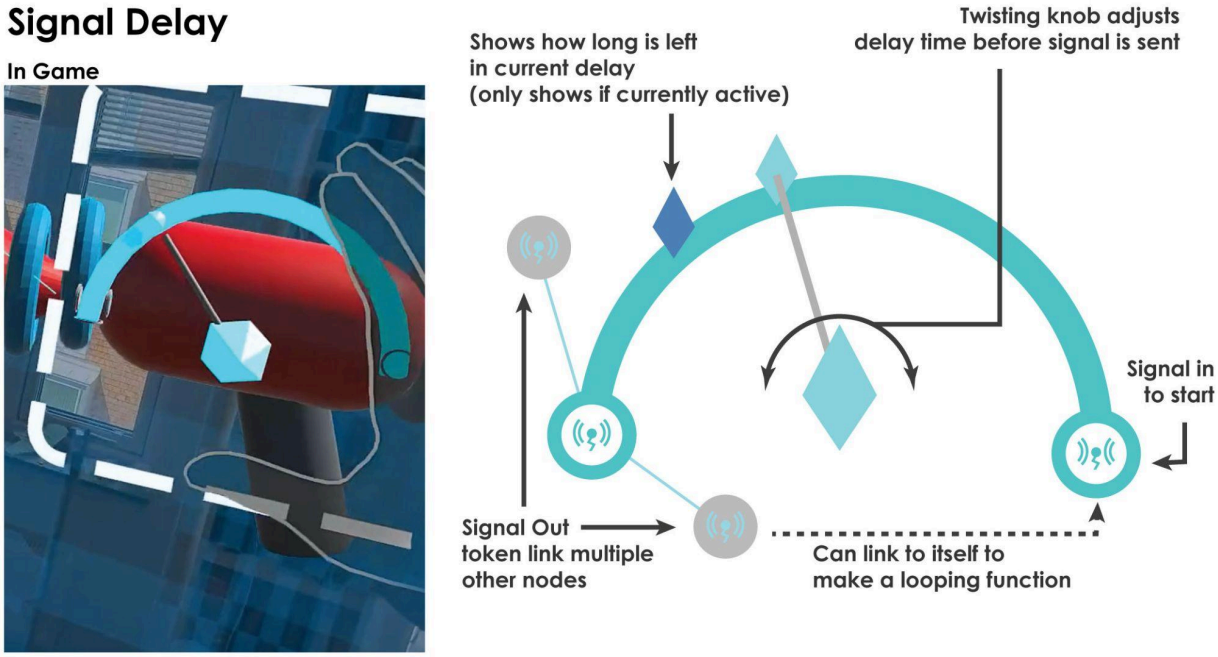


Fig 22: Signal delay waits for set time before sending the signal onward. Adjusting the knob adjusts how long the delay is. It has multiple signal out tokens and can be linked to itself to create a loop.

4.3.3 Testing and iterative prototyping

The final prototype was evaluated throughout three testing periods, each period consisting of multiple play test sessions. Players would test in groups of two. Each period was held a week apart so as to incorporate feedback into the prototype.

Each test period focused on different types of users. The first group were four IxD masters alumni. These testers claimed to be AR/VR hesitant. They had minimal experience with it but did not find it especially appealing. The second group were five of the students that made up the target audience. This group offered a great deal of insight into the role that the tool would play in their current learning. The final group were three laypeople to AR/VR development with varying levels of interest and pre-existing programming knowledge. These acted as a reality check as to how literacy may appear in laypeople.Over all eleven players were tested.

The imagined experience was cooperative building, yet it was not achievable during the project period. To simulate something similar, one participant watched a stream of the other's actions and switched to wearing the headset and built on what the other player had made.

Before playing, a brief summary of the tool’s purpose as a playful sketch environment was explained. Players were asked to pay attention to which parts felt free and playful throughout the experience. The researcher also made observations in behavior in a note book and asked follow up questions about unexpected behavior.

The players were guided through the process of building with the gesture controls and building functions. They then were allowed to build 3D objects on their own. After each participant had a chance to build objects, building components were revealed and explained. The researcher guided them through hooking up a couple alchemic nodes. Afterwards, they were allowed to play with the logic system until the system broke. Fig 23 shows some of the testers' creations.

Semi-structured interviews were held to debrief what they found playful and understandable. Inquiries were made as to what they could imagine doing with the tool if it was further developed. A sketch book containing more alchemic nodes diagrams was run through with players to see what other affordances interested them. Their responses inform multiple discussion points found throughout section 5.

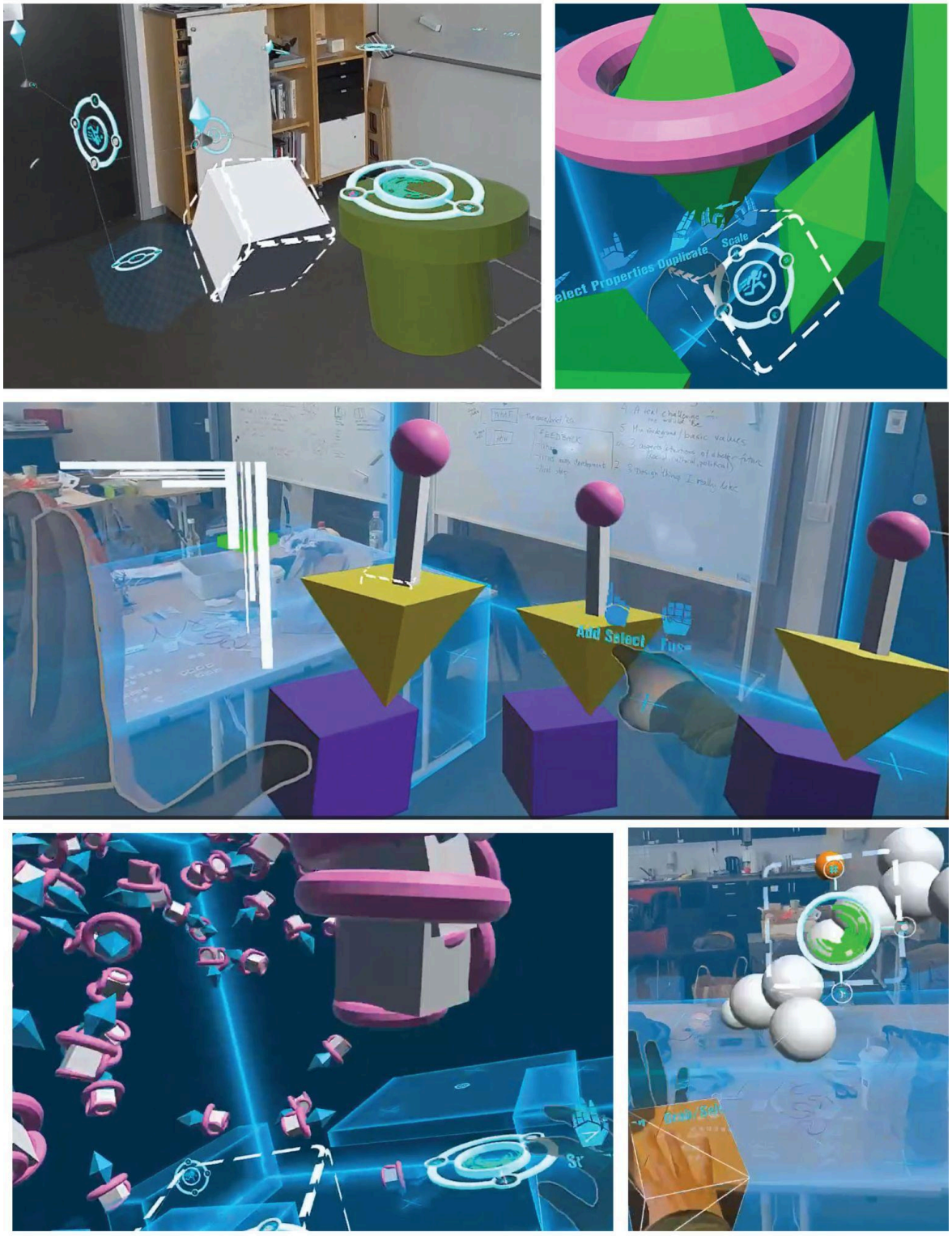


Fig 23 Play tester creations in order left to right top to bottom: a mini game that made rings appear from a pipe, a diamond monster that walk about the room by slapping the table, a series of repeating statues, a rapid statue emitter that crashed the game, a floating button that filled the room with balloon animals. Screen shot taken by researcher with permission from play testers.

5. Discussion & Evaluations

This thesis project had many layers to it. In order to engage in discussion topics that emerged from it and to evaluate specific parts, the following section is split based on design decisions. Discussion of what spurred a design decision, how it was felt by the user, and what further questions emerged from the decision are used to evaluate it.

5.1 Coupling and Literacy

ShapesXR and Unity with the Meta SDK were used by the researcher's former students' first AR/VR assignment. These tools offered a contrast between two highly different tool workflows. Nearly all surveyed students preferred authoring within the medium using immersive tools. They cited that there was immediate feedback of their edits and it was easier to "... build everything in the way it was going to be perceived." Yet the same students were frustrated with the limits to logic building within the immersive tools. Many ended up relying on group members that had prior coding or Unity experience to handle the task of logic building. They explained that this was less about an interest in playing with the capabilities of building logic, and had more to do with the approachability of the Unity setup and navigating its interface.

Wensveen et al use the lens of a feedback loop to examine digital tools that can be coupled along several aspects: time, location, direction, dynamics, modality, and expression. They suggest that "unifying action and reaction on each of these aspects makes the interaction intuitive." (Wensveen et al., 2004). Such seamless interaction could be understood as the tool providing nearly immediate literacy.

This lens could be applied to the students' experience with immersive tools on an interaction level. The tight coupling of designing for AR/VR in an immersive tool helps them quickly understand how the world will react to their actions. The Unity working environment allows for more access to capabilities of the headset, but individual creative actions are less coupled with the end result. The surveyed students seemed to all choose tighter coupling over capabilities.

The coupling lens is very similar to the affordance lens used for this thesis project. Both look at how accessible the digital world is to user action. However the material lens from phase 2 starts by looking at affordances within the digital material and looks at how the designer might manipulate it. The coupling lens starts at the tool and observes how it behaves as an intermediary between the user and digital material.

In the end the decision to take a material and affordance-focused lens gave the researcher clarity of what to look for in AR/VR in the first place. Keeping that lens in place when viewing the tools helped identify many under-exposed materials by multiple tools. In contrast, the coupling lens may have focused the researcher just on materials already accessible, possibly skipping phase 1 all together.

A future study, with more time allotted, could expand phase 2 of this project into looking across more tools and reviewing the coupling when manipulating each material. This mixture of lenses could yield a more explicit framework tailored to editing all digital materials available.

5.1.1 Balancing ease and literacy

Alchemic Reality was built on the premise of increasing AR/VR literacy by making it easier to sketch with virtual objects, logic, and spatial data. When designing a tool to make things approachable, it often removes friction that can provide teaching opportunities. It then comes to the designer to make decisions on what should be taught and where it is best to remove friction and need for technical skills.

Computer logic feels strongly connected to the technical skill of coding. Coding languages like C# are commonly used to express the coder's intent on how the computer should behave. Yet this technical skill is situated with a compiler that converts C# into binary code, which hardware uses to function. With the growing popularity of vibe-coding, using LLMs may soon be the main way to convert a creator's intent into binary code. The creator's intent works outside of the manifestation of technical skill.

The survey for former students showed a tech-based modification of their intent. Several students stated that they set out with one idea but as they ran into tools not cooperating, they had to reframe the project to something they could achieve. Their intent became framed by which affordances they could see through the tools they were able to use. Sections 2.3 and 2.6 of this thesis engage with how literacy plays a role in helping people understand what a digital medium is capable of. Literacy then also fosters intent by helping the creator imagine new possibilities.

Alchemic reality aimed at literacy of specific affordances of AR/VR above technical skills. It situates itself at the place where players can foster broader intent through play. Final play testers' behavior demonstrated that it does. One of the laypeople excitedly stated that "[they] never thought [their] lamp could act as a controller" explaining how they would link it to some of the potential future alchemic nodes. A former student called it a "Microbit for augmented reality", Microbit being what they used to first learn electronics. They continued "if you added logic gates I would be linking things up for days."

The narrowness of the logic blocks provided is the prototype's biggest weakness. Time only permitted for development of a handful of potential blocks. There were enough to examine the playful nature of the interaction with the logic, however it railroaded them into thinking specifically in terms of the logic provided. The players were still unaware of the many unincluded affordances a dream tool could provide and none of the play testers' imaginations ventured beyond the scope of what had been shown to them.

5.1.2 Ideating versus implementing

During phase 2, working with VibeCodeXR was very different from the other tools. It has a semi immersive workflow, which is as follows: convert intent into verbal language> prompt the LLM > wait for the LLM to build > test via a generated link on your headset > repeat until finished.This process looks more like a rapid prototyping type of flow, with a focus on the implementation of an idea, perhaps already developed before engaging with the tool. Each action costs the creator tokens, meaning using it for rapid prototyping can get costly if the creator doesn't already know what they are looking for. Ideating has an associated cost in the form of these tokens.

During one test session, the researcher intended to use the tool to sketch an AR experience of the Acropolis in Athens. The intent was to let users learn about what was happening there at different times in history. The idea was loose and the researcher wanted to move back and forth between what the experience could offer and which different modes of interaction should be used. This was an inefficient use of tokens but the prototype itself also started to run into problems. In the code, old systems were merged with new ideas. Eventually this led to the prototype becoming too unstable to test and the researcher had to start over.

In another test session, the idea was sketched on paper first and explored in a more traditional way. Afterwards, a game design document was made for a city builder with an advanced UI. Uploading a full design document at the beginning made for a much cleaner experience in VibeCodingXR. In talking to two other senior developers that had more experience with vibe coding, this seemed to be an optimal work flow in other LLM based coding tools as well.

As the city builder project was developed further it became clear that the UI laid out in the design document didn't register well in the headset. Iteratively, experimenting with other solutions led to an unstable prototype again. Personally debugging it was difficult for the researcher. Even with knowledge of the coding language, the coding style of the LLM was hard to follow.
VibecodeXR is still in beta as of writing this thesis. If it is developed further many of these problems with prototypes may be fixed. However, even as these tools become better at implementing code, there are important parts of sketching and ideating that do not come so naturally to the verbal interface of an LLM. This led the project to situate itself as primarily a sketching tool that would allow for students to play with the interactive and experiential elements of AR/VR.

The experience also led the researcher to ask playtesters how they saw experiential tools like Alchemic Reality situated next to more instantaneous implementation tools found through LLMS. Responses to this are discussed in section 5.5

5.2 Using a Gesture Based Interface System

Alchemic Reality uses a gesture based system as the main interface with the system. This decision was to explore ways in which AR/VR can work outside of the traditional 2D interfacing techniques. It also encourages the player to start thinking about the presence of their hands in AR/VR.

A reactive system of icons float above the wrists of the user and updates to show which commands are available for that hand. When a pose changes the state machine, larger icons icons float near the user's fingers, see Fig 24. Sounds were also played.

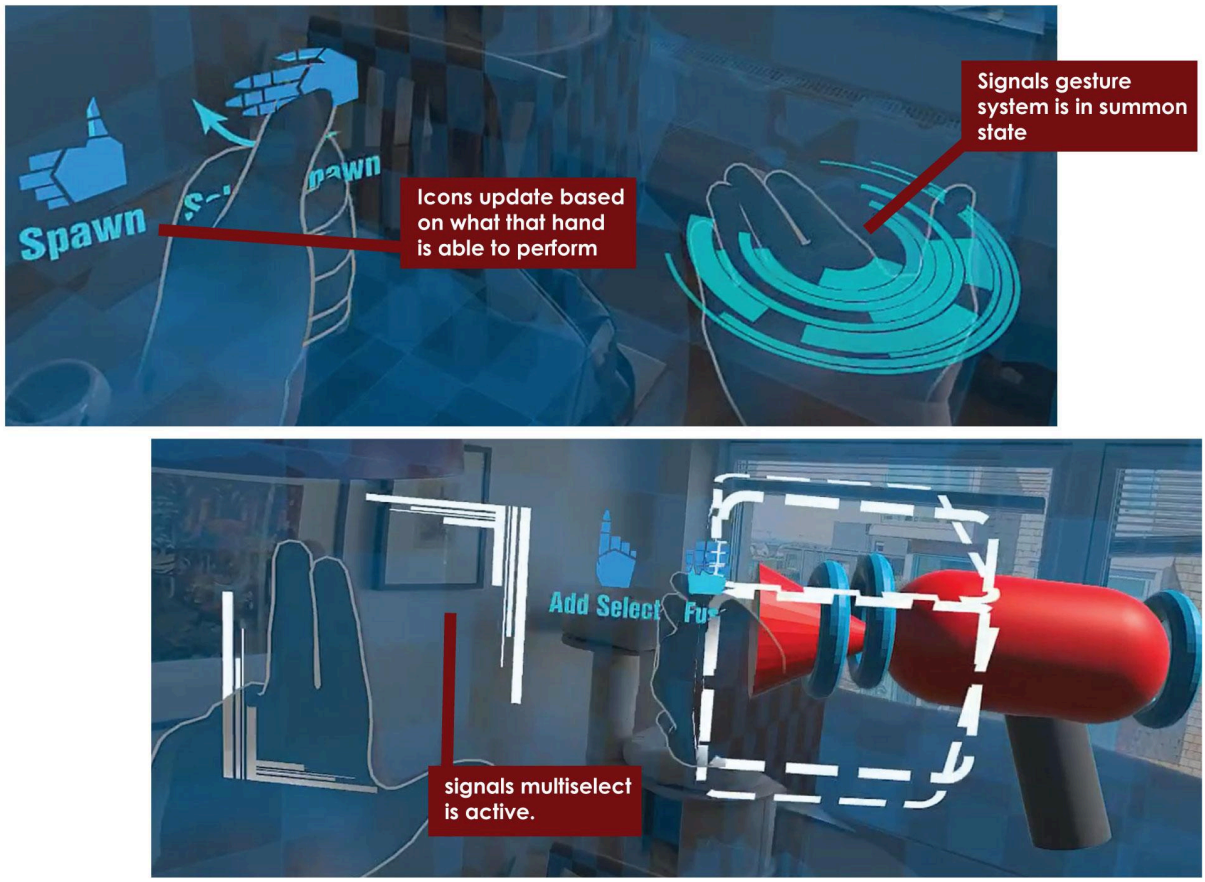


Fig 24: Icons were accompanied with a unique sound to reinforce feedback. Graphic made by the researcher.

Using both audio and visual feedback to signal a command was being registered came from player feedback. Early testers showed that at first, the necessity of sustaining the pose was difficult to remember. Adding more feedback made them more aware of the system's current state.

Play testers saw gestures as an intuitive way of building. It took little instruction for users to quickly start building all manner of objects. They also commented on how they preferred this to controller- or mouse-heavy 3D tools they had experienced in the past. There are improvements

to be made to precision in building. Players often had to readjust objects after placing them. This was due to the system not detecting that the player had let go even when they thought they had. This even happened when players moved very slowly and intentionally, as their hands would slightly twist and rotate as they let go of their grip. More tuning or tool modes would need to be developed to match the precision of a tool like ShapesXR's placement.

2D iconography for spatial hand poses presents a problem; they served as good reminders but had difficulty conveying initial hand poses without further instruction. Users commented that after some time, they noticed that different colors were used for palms and the back of the hand, but even then the facing of the hand was difficult to recognize. A 3D icon system could assist with this in future iterations.

Players expressed that the 2D window was still useful in comparison with the gesture based commands. Users enjoyed the gestures for many commands but as one tester described, the 2D menu felt "safe and familiar". This same tester said they would not enjoy it if the 2D menu replaced the gesture system, but expressed that they liked having both work together for different tasks. For example, it was easier for them to explore properties that had a lot of options when the options were listed next to each other.

Placement of the 2D menu became a bit of a problem. When they opened the menu it appeared right next to their hand. Players would hit buttons unintentionally right as it opened. One tester asked if they could connect the menu to their left hand. This could be explored more deeply in future iterations.

5.2.1 Non uniformity of the human hand

The Meta Quest has limitations when it comes to detection. Machine learning is used to validate detection of the thumb, index and middle fingers. This makes poses with these fingers much more reliable. Pinky and ring finger pose is detectable but inconsistent. Since human hands are not all the same, this can produce complications if important commands are tied to less reliable finger formations.

While machine learning can help with recognizing a finger pose, it can not help with the user's physical abilities. Among all testers throughout the process, different poses were hard to achieve per user and per hand. There were several poses that detected abduction, making a V with two fingers, these were well detected, but it was hard for some users to perform it. Even if they could do it on one hand, they may not be able to with the other.

This makes a deep gesture system with a wide variety of hand poses difficult to achieve. It means that testing hand poses with users is imperative for usability studies. Designers need to choose attainable hand poses for their audience. There is an obvious problem for the system if the user is missing fingers important for input commands.

Future iteration would need multimodal input. While this prototype does not include extra options to initiate commands, ideally any gesture-based input system would be coupled with a secondary or even tertiary input alternative to the same command.

5.2.2 Bodily awareness and the reality of accidental input

Gesture detection on the Meta Quest relies on the camera positioned around the headset. The detection range is fairly wide around the user but does have blindspots. When the system cannot see a hand, it tries to guess based on the last detected information. These guesses by the system can disrupt the designed command system by sending the wrong information. Users are not always aware of this happening. This was especially observed when a pose is being held with one hand, while the user focuses on doing a pose with the second hand. This is why the use of multisensory cues tied to different actions are important. If the cues are visual only, the user can very well not be looking and lose context for how the system is reading their input. Visual cues that follow the user's view such as a Heads Up Display (HUD) are a way to create visual feedback, but risk cluttering the view of the user.

The same detection also creates user generated errors as users are not always aware of how they are positioning their hands. If a command is tied to a neutral position of the hand, the system may detect the pose when the user is not paying attention. This means that it is important to pick poses that must be intentionally made or are harder to accidentally make. This must also be balanced with choosing poses that are easy enough for your user to perform as mentioned in section 5.2.1.

Using hand direction as a secondary requirement to pose was used often in Alchemic Reality to midgate accidental input. This could even allow for the same pose to give different commands in future iterations.

5.3 Visualizing Logic

The decision to make a consistent language to the visual logic helped with denoting an affordance in the system. For example, spatial data is not visualized or engaged with often in immersive tools. Alchemic nodes are attached to each piece of furniture in the room. They are the same type of alchemic node used for collision detection and can be connected. When users turn on the app, they instantly are informed that these pieces of furniture can interact with the system. In this way the circles act as a seam, showing possible places to repurpose the system.

Coding and logic building more often happen outside of immersive experiences, even in heavily immersive tools. For example, FigmenXR allows for building assets and setting scenes up while immersed, but any coding of interaction is done outside on a traditional 2D screen.

RecRoom features an extensive tool set of immersive logic building blocks. Reading user reviews of the system showed that many actually enjoyed the physicality of logic building, for a time. Most veteran builders seemed to end up using the desktop editor of the tool. However, this

could be because the interface in both desktop and immersed versions was designed with a 2D interface organizing all parts into tabs and text fields.

While building in the tool, the researcher found themself often wishing they had a mouse and keyboard at any point where a floating 2D menu appeared. As many aspects of the logic building relied upon these menus to function, the normal mode of interacting–a mouse or keyboard–felt more natural. In contrast, when building with parts of the tool that relied less on these menus, for example connecting a rotator to a door hinge by literally moving it with your hand, felt much more enjoyable. This experience inspired the decision to make a node based system that relied less on menus to connect and manipulate.

5.3.1 Readability

For the system to be playful, it needed to be quickly readable as mentioned throughout this thesis. All testers needed instruction on how to read the alchemic nodes but once they learned one of them, they read new circles faster each time. Several of the testers began improvising with the other circles they found immediately after learning about only one.

The most-needed guidance centered around what the requirements were for each node beyond a signal in. The iconography was not quickly understood, and in cases where a node took two float inputs, it was not telegraphed what each float did specifically without some experimentation.

The delay node was the most difficult for users. It was the only one that featured multiple outputs and this led to some confusion as to what the use was. This was met with inconsistency in the way the rotating knob moved when grabbed. The testers gravitated toward dragging the crystal around rather than twisting it. Even after explaining that a twist was needed, it didn't seem to resonate or be understood by testers.

5.3.2 Guiding User Action

As mentioned in other sections of this thesis, many design decisions revolved around signaling to the player where they can act. Much of this framing was first investigated based on responses in the expert interviews.

The experts were asked to explain why they chose to make their application in AR/VR rather than other mediums they were familiar with. This provided a lot of insight into what each found special about AR/VR. Immersion was mentioned often, the ability to really focus the user on an experience through multisensorial designs. Yet even when immersed, users' attention isn’t guaranteed.

Experts also emphasized the importance of guiding the user toward actions. While immersive experiences can block outside distraction, the experience itself is a wide canvas which allows

for the audience to go in unexpected directions. For one designer, this focus was added through audio narration asking them to act at different points. For another designer, presenting a UI helped them navigate different options. Multiple experts expressed that some kind of active telegraphing or heavy handed guidance was key.

One expert interviewed is a creator of educational experiences in Minecraft, and they pointed to a more ambiguous type of guidance. A main reason they liked building in Minecraft is that their target audience already had gained a certain understanding of how the game world worked before getting into the designer's educational experiences. The expert leverages that implicit knowledge as a type of guidance to action.

They went on to explain that the open design of Minecraft, discussed in section 2.7.1, played a role in their own journey becoming a designer. The game creates an actual starting point to explore by providing a digital world to play through. The expert pointed at this as a gateway into modifying the game. The users are not asked to invent something out of nothing. It gives some suggestions of places to start working and even some in-game challenges, but doesn't exactly dictate what to make or how to solve any of the problems players encounter. This lets the players rely on their own creativity. They are quickly led to see it as a world they are allowed to alter. The expert said it eventually becomes second nature to “think” in the medium of Minecraft.

Zelda that is discussed in section 2.7.1 demonstrates a similar idea to the experts observations of Minecraft's invitation to be creative. In both cases the game play is focused on the player to act creatively to solve challenges with the system and in turn teaches the player to stretch its limits or even appropriate it for another use. These games in part inspired Alchemic Reality to foster AR/VR literacy through play. Though, the concept could be taken further in Alchemic Reality. In future iterations, making in-game challenges could play a role in inviting the users to engage with the alchemic nodes and learn how they work.

5.4 Visualizing Spatial Data

The first playtest session revealed that the prototype had almost fallen into a trap. One of the testers mentioned it was fun but they, “just didn’t see where this fits into [their] life if [they] don’t play games.”

There was a common trend in the reviewed AR/VR tools. Many advertised that they offered AR capabilities. What this really meant was that you could enable the visuals from the passthrough camera. This led the researcher to question if this really was AR. Is the camera data of the physical world all that is needed?

During the first test passthrough was enabled so one could see the world. The researcher tried to bring the real world in even more by adding visuals to the virtual model of the spatial data. Playtesters even had a lot of fun playing with the fact that the physics engine considered the spatial data, allowing them to bounce virtual objects into it. Yet this was not far enough to expose the connection between the virtual and physical worlds. During this test the only

alchemic nodes available were a summoning, a delay, a float and a collision node. All of these nodes focused on affecting things in the virtual world.

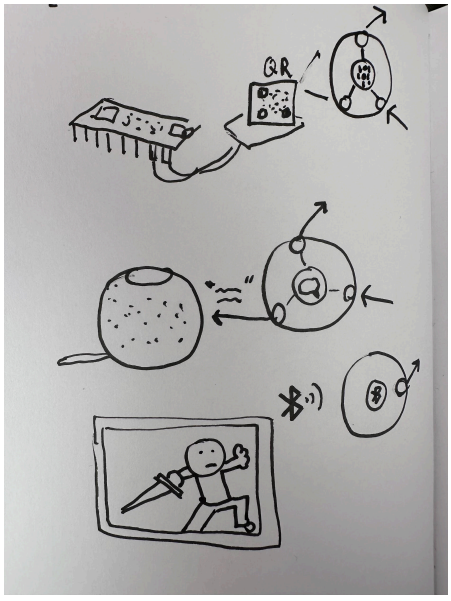
It was easier to begin developing for the virtual world as that is where Unity is situated, it is a virtual world-building tool. Virtual tunnel vision is a default reaction to staying within a tool that only recognizes the world after code is tested.

There could be something to be learned from the *mobile first* web design in the early 2010s. Phones were rising to be the primary method of engagement with the internet. Web designers were used to designing for a wide screen desktop, adding JavaScript to shrink the content layout for phones after the fact. Mobile first design was a paradigm shift that asked designers to design for the phone first, in order to ensure that the system was designed with the mobile phone's unique affordances (Wroblewski, 2011). The author of this thesis was a designer at the time and remembers how quickly web standards changed when designers started reframing their outlook.

This thesis has been using the term AR/VR in this order to reflect a potential need for an AR first paradigm. If the real world is taken into consideration first, there is a higher likelihood that designers can find new ways to make the real world more present in experiences.

The design response was to add alchemic nodes to all environmental objects in the spatial data by default. Each prince of furniture in the environment was also given a tag in the system. This way objects could even track which parts of the environment would trigger logic blocks.

Playtests after this addition saw users engaging much more with ideas of how to use their environment to link into the system. Though it could be pushed even further. Sketches were made to explore more physical world objects, see Fig 25. These were shown to the final play testers along with other potential alchemic nodes. The implications of integrating the physical world became one of these laypeople's main interests. They said they would use these advanced functionalities to wire their smart devices to non smart objects in their home.



This did not resonate with one of the testers. They said that for them seeing the spatial data was off putting. They observed how the visualization made them see how inaccurate the spatial model was, exclaiming that, "it shows that chair as a box which makes me immediately dismiss it." This is taken as a different benefit of the visualization, the user is becoming more literate of how the system views their space and what errors it is prone to.

Fig 25: sketches of potential real world nodes to add. Qr code based bluetooth connector that sends serial codes to a micro controller. A text command that could be sent to a smart home device. A connector to smart tvs for potential video streaming.

5.5 A place for playful spontaneous creativity.

Where a tool is situated in the design process is an important part of its identity. Section 5.1.2 examined how some tools might be better for implementation while others are better for ideation. Alchemic Reality was designed as a gateway into the medium, situated at the beginning of a designer's journey, and aims at fostering *playful spontaneous creativity*.

Testers that considered themselves laypeople or artists, expressed they could see Alchemic Reality as a way to better wrap their head around how logic plays into the spatiality of AR/VR experience. One of the former students said,”I would rather have used something like this in our class”. This student expressed that it would have let them engage top level logic construction. Having logic in this tangible way also let them learn about concepts in a way that was more familiar to them. One tester said that as an artist, ”I learn with my hands first.”

Amongst the playtesters were a few who considered themselves software engineers. They expressed that they enjoyed playing with logic in this new way but could not see themselves creating full-fledged large programing tasks. One such test user expressed that they “would still see [themselves] using it though.” For them it seemed like an easier way to perceive how logic applies differently in a spatial context. They imagined a workflow that started in something like Alchemic Reality to discover top level concepts of an app that then would be used to design a spatial app using a more flushed out coding tool.

Several testers said they would be excited to see it as a more flushed out AR sketch pad, with more affordances opened up through the alchemic node system. Testers were asked if this held any value compared to their current methods of sketching such as body storming or pencil and paper. The student and laypeople testers all said they found value in how it telegraphed possibilities within the AR/VR context.

Testers especially liked that they could send signals from their furniture. This was an aspect they had not thought about until that point. Therein lies both the boon and bane of the tool. It reveals seams but it is limited to how many seams are built by the tool designer. These testers did not detect the myriad of affordances in the list from phase 1 that could have been tapped but were not.

For Alchemic Reality to become a truly playful spontaneous creative environment it would need to expose many more affordances across more of the material categories. Not just to make more individual functionalities but to see what unexpected connections the player could make with the many AR/VR materials working together.

5.7 Bias feedback from testing with your own students

As mentioned, some of the testers were former students. There is a lot of bias that can happen due to this relationship. To help navigate this, two useful mechanisms in the research were implemented. The first mechanism was sourcing an outside audience for contrast in answers.

The second mechanism was to ensure interviews with students were structured around critique of the project. Instead of questions such as ‘what did you find playful in the experience?’ They were phrased as ‘where was the most friction in experimenting with the node system?’ to better solicit constructive critiques.

Additionally it became more important to note behaviors during the test that could be inquired about in the interview. ‘I noticed you repeatedly trying to click the shape option, what was happening there?’ Questions in this form yielded more insight than those that left it open to talk the project up. Along with the critical feedback the researcher sought, the students intermixed more positive insights of their own accord. This made for more constructive conversations overall.

6. Conclusion

6.1 Knowledge Contribution

This thesis has produced several knowledge contributions as a response to research questions.

R1: If we consider AR/VR to be a medium made of multiple materials to be manipulated – what are the available materials?
The project produced a preliminary list of materials and affordances found within AR/VR when using the Meta Quest 3, found in appendix B. This list could be further developed and vetted in future research projects.

R2: How do different AR/VR authorship tools give users access to these materials and which materials are locked behind technical skill walls?

The project found that this is hard to directly answer, as it is highly dependent on the tool. A review criteria connected to the material list, found in appendix C, was made to be applied to tools that build for the Meta Quest 3. These criteria also could be further developed under the lens of digital coupling.

R3: What design choices in AR/VR authorship tools lead to deeper literacy for laypeople and new designers?

This project found that AR/VR literacy deepens in laypeople when they are allowed to play with more of the affordances found in the medium. Decisions on the tool designers end dictate what affordances are exposed. The discussion of this thesis, in section 5, outlines several decisions that foster literacy including interfaces, visualization methods, and exposure through play.

How do AR/VR authorship tools need to evolve to allow for playful spontaneous creativity amongst new designers and laypeople?

Ultimately this thesis found that there is a large gap open for playful spontaneous creativity tools in AR/VR. Most AR/VR authorship tools are built with implementation in mind. Immersive tools that allow for sketching often lack many of the materials that make up AR/VR. Alchemic Reality sits as a core knowledge contribution of this thesis. It exhibits an approach that makes several under-exposed materials more accessible to laypeople. New alchemic nodes can be developed for the system as a way to both offer access to additional affordances as well as telegraph to players that an affordance exists.

6.2 Final Perspectives

This thesis project set out in search of playful spontaneous creativity in AR/VR tools. The resulting research gives insight into a need for designers to expose laypersons to more of the many affordances found within the medium. Doing so invites novel ways in which the technology might situate in their lives.

It is the view of this thesis that it is not the technological capabilities of HMDs, nor the community of AR/VR design enthusiasts, who gatekeep the craft; it is the tools.

Playful spontaneous creativity tools like Alchemic Reality show promise in helping players gain more AR/VR literacy. Exposure to more materials and affordances in AR/VR could be given through alchemic nodes, which act as seams. All current affordances are exposed in traditional code; it still remains to be seen if this visually-based system could ever attain the same level.

LLMs are providing faster implementation of ideas as advancements make them more accurate translations of intent. Even in light of these advancements, it is important that we do not forget the role that playful spontaneous creativity has in helping us design the *right thing*.

7. Appendix

A: Video demo of prototype - https://youtu.be/l6GDh5_KMYk
B: Glossary of material mapping
C: Material Inquiry Questionnaire

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Appendix B: AR/VR Material Glossary

This glossary is written to understand materials of augmented and virtual reality (AR/VR). AR/VR straddles a line between virtual and physical worlds. Some materials exist only in one of the worlds while others are represented in both.

It is sectioned into overarching general sections that group materials. These overarching sections are mostly based on Marika Hedemy's understanding of the “strands” of mixed reality (Hedemyr, 2023). The main difference here being that storytelling and over time categories are separate.

The material is explained by how designers might manipulate affordances in the material through specific design choices..
Some materials are not directly manipulated by the designer, such as a user's imagination. These materials can be leveraged in AR/VR if the designer makes openings for it.

User Body

These materials have to do with how the user's body blends with the digital experience. They also have to do with the user's presence and how they can express themselves in the AR/VR experience.

Gesture

This can also be described as physical expressions of the user's bodies. Designers can choose how bodily movements and poses in the physical world are translated or coupled with actions that affect the digital world.

Mobility

This describes how much users navigate the AR/VR experience and kinaesthetics. Designers can choose to make the user remain stationary while the rest of the experience moves around them or they may invite the user to move from place to place. Users' ability to move could be of concern.

Memory

This material exists outside of the designers ability to control. Designers can provide opportunities for users to draw upon their memories to define the AR/VR experience. This is prevalent in VR therapy where users are asked to visualize things from their past as they drew and moved their bodies through virtual space.

Creativity

This material exists outside of the designers direct control. Designers can decide to build parts of the AR/VR experience be editable and open for novel retooling or expression. This allows for users to channel their creative thinking into the experience.

Group Size

This denotes how many users the AR/VR experience is built for. Co-drawing in AR/VR, for example, requires two people to function. Multiple expert practitioners noted that this aspect affected users' stamina. When users are together in an experience they typically engage for longer.

Tacit Knowledge

This material exists outside of the designers direct control. It is the knowledge that comes from common experiences. Designers may choose to link story telling or system interactions to these common experiences. For example using the gestures of an air guitar to make the system play a sound.

Stamina

This material is outside of the designer's direct control. It pertains to how long the user engages with the VR/AR experience. This is affected by decisions about what kinds of physical movement are encouraged, the amount of social engagement they feel. It also has to do with the individual user's physical felt reactions to designers choices of stimuli produced by the experience.

Site/Environment

In AR/VR, it is up to the designer to choose how tightly coupled the physical and digital worlds are. While VR primarily engages with a fully immersive digital world, AR balances both in different degrees. As such, these materials were separated by which world they live in. Depending on the design, the physical world is altered by the designer such as in AR/VR art installations, yet other AR/VR experiences simply react to the physical environment by feeding data the digital twin of the space or even ignore the physical space altogether.

Physical Spatial Shape

This is the actual contours and shape of the space the user is in during the AR/VR experience. Designers can decide how to shape or frame the space. For example, are users in a crammed room with furniture or an open free space?

Physical GeoLocation

This has to do with where on the earth the experience is situated. Designers can choose where the AR/VR experience is accessible from. They can also make decisions based on the social context of the place. For example, an AR tour of an old canal that can only be accessed from a certain spot so that the experience can play with the social meaning of that specific canal.

Physical Objects Present

This denotes the presence of specific physical objects and their role in the experience. In many modern head mounted displays these objects can be detected and identified by the virtual system. Designers may include specific objects in the environment that are meant to be used in coordination with the system. A special chair that plays a part in the story telling for example.

Spatial Digital Twin

This is the spatial model of the room as stored or detected by the AR/VR system. In Many modern HMDs this data is scanned before the experience and leveraged while some AR/VR experience regularly scan and refine the spatial model of the environment. Designers may use this in concert with other materials to create AR/VR experiences that adapt and change based on the physical spatial space.

Tele-location

This reference to the connective capabilities of modern HMDs. Designers can include other users from remote locations within the experience. They can also decide what information is sent or stays per location.

Digital Proximity

This is the position of objects relative to each other. Designers can decide to use this information to dictate different interaction types. For example, voice chat in a remote experience that becomes quieter as users get farther apart from each other in the virtual world.

Technology/Media

Which technologic devices present are a design decision. For example, which HMD is the user wearing. Within a given device the individual capabilities it has manifests in its sensors and actuators each represent a potential material to utilize. For the purposes of this list, the section is seen through the lens of what a Meta Quest 3 is capable of on a surface level.

Display

These are LCD displays for the Meta Quest 3. AR on HMDs is unique in that the real world you see is actually a display of the real world, referred to as passthrough data. There is a processing of the passthrough data before it reaches the displays thus unique affordances exist in how the real world is displayed. Designers can play with what is or is not shown on the displays or how that data gets rendered.

Camera

As discussed in the display material the headset is equipped with a camera to capture images of the world around the user. Designers can use this data in many ways, such as in computer vision, to create different processes other than just sending it through to the display.

Depth Sensors

The Meta quest uses depth sensors to build spatial data. Meta's recommended use of this sensor is to use it outside of an experience and create a durable data mapping of your room that is then accessed later within an app or experience. However the sensor is still accessible to technically advanced designers and can feed data for processing at whatever time or in whichever way the designer sees fit.

Controllers

The Meta Quest 3 has the option of handheld controllers as a means of user input. Along with buttons and gyroscope input data they can provide haptic feedback. The designer must choose how much that controller comes into play.

Gesture Detection

The Meta Quest 3 detects the position of the hands and some other body part position data. Designers can use this to create unique input via finger and hand movements. How much is tracked and translated into the digital space is up to the designer as mentioned in the User Body section.

Computer Vision

As mentioned in the camera section computer vision software is actively used on the Meta Quest 3 and they provide libraries of data to help identify what objects are in front of the camera. Designers are able to use this data as they see fit.

CPU/GPU

The processing power of the Meta Quest 3 has its limits. Designers must make decisions along the lines of how much stress they want to put it under. Similarly, AR/VR designers make choices to tell the user how much stress the CPU/GPU is under. Designers can decide whether they want to have the experience processed on the headset or streamed from a connected computer.

Speaker

The meta quest has speakers built in. Designers can choose to attach other speakers to the headset via bluetooth or wired connection for a range of sound mixing results. They are also subject to the user's control of the volume level of the speakers.

Microphone

A built-in microphone allows for the recognition or recording of sound from the user or environment. Designers can translate recorded audio into their own system for processing in novel ways, such as voice command using a large language model

Large Language Model (LLM) Interfacing

Meta has built LLM tools specifically to be used with the Meta Quest. Designers can implement this as a processing layer in their experiences. They are subject to LLM token usage limits.

Streamed Video

Streaming media via wifi is a highly utilized feature in many experiences on the Meta Quest 3. Even if the designer wants the experience to stay on the headset they may still choose to leverage cloud data to be streamed to the head set via its connections to networks.

Story Telling

These materials have to do with the story telling mechanisms of the experience. Designers express story telling in a myriad of ways. From interviews and surveys there were several common types of story telling that used specific digital asset types within AR/VR.

Graphics

Designers may include 3D objects or 2D graphics to tell stories. In AR/VR this visual story telling is highly utilized, some experience being almost purely graphical.

Text

Text prompts are used to tell stories to those able to read. Messages and system prompts are also often displayed through text.

Audio

Designers often prerecord sound to be used in novel ways in AR/VR. Sound can be spatialized in these experiences making a sound feel like it comes from a certain direction. It can also be filtered through all types of processing to make new and novel sounds. Sound effects are used as feedback for users' interaction in the world. Lastly it can be used as verbal guides and cues within the experience for story telling.

Haptics

When using the controllers designers can make haptic feedback in the hands to give textural or felt storytelling. These can be programmed per controller for highly nuanced expressions of the digital world.

Co-creation

Multiple users within an experience are not by default given the same control over the environment. Designers can allow different users different levels of control in the digital world, assigning roles. This even extends to instances where the designer invites the user to modify and build the experience such as in VR drawing.

User Interface

User interfaces denote the system by which users give commands to the system. Some experiences want the user to simply watch with minimal interaction where others make deep interfaces that offer a lot of possible interactions with the world. The way this interface manifests – its input types, what it computes, the feedback it gives, etc – is up to the designer to create. User interfaces are also strongly connected to the over time category.

Over Time

AR/VR experiences are rarely if ever static and often change over time. This allows for opportunities to design interactivity and novel experiential worlds. This is not unique to AR/VR. For some of the interviewed experts, being able to wrap the user in layers of processes that happen over time was what drew them to build in AR/VR. Students especially were interested in experimenting with over time interactions between the user and the system.

Code Logic

Can also be understood as mathematical or computational logic. Designers must decide what functions the computer performs and how they weave together to make the final experience. The logic is often modular meaning functions can happen in many different orders to create novel experiences.

Animation

How objects move and change over time. Designers can move a whole object from point a to point b along a path. They can also make an object change its shape or switch between visual states over time.

Time Tracking

Designers can decide the timing of specific functions. Time can be tracked relative to the start of the experience or coupled with the real world time, date, season, etc. Time tracking can be represented in clear or ambiguous ways in the experience.

Virtual World Reactivity

Designers must implement any reaction the virtual world has to the user's presence or activities. The virtual world can calculate and react in novel ways beyond the scope of the rules that govern our physical world.

Virtual World Passively Updating

Designers must choose which rules passively govern the virtual world’s behavior, such as physics and entropy of virtual world properties. The world may also evolve or change and transition with no input from the user such as scene changes in immersive video.

Physical World Reactivity

The physical world may react to the user in concert with the virtual system. This often takes broader design approaches of integrating physical systems that the headset can talk to. These techniques can be seen in installation art pieces.

Physical World Passively Updating

The rigid material of the real world has rules that govern it. Designers must design to this reality’s rules even if they have fully immersed the user in a virtual world. For example if the user must move around the room they eventually will hit a wall or if they jump gravity will pull them down in the real world. There are gadgets and tech that try to manipulate these rules to a degree such as omnidirectional treadmills. At the point of time this research is being done much

of these passive world updates can be designed to but are outside of the capabilities of Meta Quest 3 to directly manipulate.

Captured User Data

Designers may put parts of the system in place to track and aggregate users' choices over time as a form of meta data. This data was seen to play out as feedback within the experience. Fro example, what rooms a guest of a virtual conference has visited. It also is used for reports used by designers to analyze user behavior.

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